Cubic Time Dimensions in Quantum Mechanics, Finding Electron Orbital States with Time

Second edition, revised: proper time, coordinate time, and the periodic table

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Abstract

This paper models atomic structure with time, not space, as the primary variable. This second edition keeps the first edition's premise, that an electron's states can be found from temporal interference, and replaces its fitted three-time model with a construction that has no per-atom parameters. Two clocks are used. The electron's own clock is its Compton phase, and a stationary state's clock is its energy phase. The Dirac virial theorem gives $E = mc^2\langle\beta\rangle$ exactly for inverse-square forces, and only for them. Reading $\langle\beta\rangle$ as the electron's average clock rate gives the premise, $E = mc^2\langle d\tau/dt\rangle$. Classically the same statement is Landau and Lifshitz's relativistic virial theorem; the theorems are known, and what is new is their reading as clock rates, carried through every element.

- *One-electron atoms.* The construction reproduces the reference ionization energies of H and He^+, and Herzberg's deuterium Lyman-α line, up to the Lamb shift. The residuals are $+2.48$, $+8.18$ and -3.26 ppm, each within 0.08 ppm of the measured or QED-scaled Lamb shift.
- *X-ray lines.* With the historical screening constants, the Moseley RMS error falls from 5.7% to 1.05%. The Kα doublet, which is exactly zero without relativity, comes out within 0.66% for $Z \geq 40$.
- *All 118 elements.* In any mean-field atom whose interactions are all inverse-square, the total energy is the sum of the electrons' proper-time deficits plus a nuclear-volume term of at most 0.6%. The identity closes to 1.1×10^{-7} in self-consistent Dirac–Slater atoms for all 118 elements. Each orbital energy splits into a *clock* term, a *shelter* term from the charge lying farther out, and an exchange term. With no per-atom parameters, first ionization energies agree with the 99 NIST reference values to 5.0% RMS (median 3.2%), and K-shell binding energies from neon to uranium agree to 0.46%.
- *Decay clocks.* Spontaneous emission obeys $\tau = 3T^3/(32\pi^3\alpha\, t_d^2)$, a cube of the transition period. It gives hydrogen's 2p lifetime to 0.1% and the alkali resonance lifetimes to within about 8%. Alpha decay follows the Sommerfeld ratio $\eta = Z_1 Z_2 \alpha/\beta$. With the textbook nuclear radius and no fitted constant, 35 half-lives spanning 29.8 decades are reproduced to 0.55 decades RMS. The scatter about a single offset depends only weakly on the radius (0.37–0.46 decades); the overall level does not.
- *The premise is now a derivation.* By the Hellmann–Feynman theorem, a known route for the Coulomb field, the electron's average clock rate is the response of its energy to its rest energy, $\langle\beta\rangle = \partial E/\partial(mc^2)$, for any potential. The premise then says the energy is proportional to the rest energy, which holds exactly when the only coupling is the pure number $Z\alpha$: the inverse-square force.
- *Dark energy.* The framework's static temporal volume is the cosmological constant. A pre-registered test of the smallest change it allows, one time dimension switching on, finds no

support in DESI DR2 (Sections 7.4 and 8.9).

- *The atom tests the framework's own cubic law.* Section 7 measures the time-dilation exponent k directly from 14 hydrogen-like ions' spectra: $k = 0.49996 \pm 0.00010$, matching special relativity's $k = \frac{1}{2}$ and excluding the framework's originally proposed $k = \frac{1}{3}$ by about 1600σ. This is an independent check inside bound states; the tightest direct test, with ions stored at $0.338c$, bounds $|k - \frac{1}{2}| < 1.9 \times 10^{-8}$.
- *Where cubic time lives* (Section 8). On the satellites named in the framework's own GNSS proposal the cube-root clock law fails (Galileo 5 and 6: about $27\,000\sigma$). A cube of time does contribute, to radiation and orbits: mass is a cubic time, $GM/c^3 = (a/c)^3(2\pi/P)^2$; the radiation-reaction force is the jerk, $m\tau_0\, d^3x/dt^3$ with $\tau_0 = \frac{2}{3}\alpha t_C$; light from a bent path arrives at a time cubic in its emission time; and in the quantum limit the power an electron radiates grows as $(1-\beta^2)^{-1/3}$, the framework's exponent, though for unrelated reasons.

The first edition's starting time t_1 turns out to be the electron's Compton clock. **This revision adds a worked numerical example after every equation**; Appendices C and D give the constants and the scripts behind every number.

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Provenance tags. Each tag marks the status of the claim it follows.

- [FORCED-BY-DEFINITION]: follows from a definition or identification.
- [FORCED-BY-MECHANICS]: follows from the stated equations of motion with no further choice.

- [MEASURED]: a literature value, used only for grading and never fed back into a calculation.
- [ASSUMED]: a stated modelling choice.
- [FITTED]: adjusted to data.
- [OPEN]: a question raised and not settled.

The paper refers to equations of the first edition as "1st-ed. eq. (n)".

1 Introduction

Quantum mechanics has traditionally been built on spatial coordinates and wavefunctions. It gives a robust account of particles at the microscopic scale, but it offers little temporal labelling of the atom. Hydrogen is rarely written in seconds per radian, and we are not aware of a standard temporal description of the periodic table.

The first edition started from a practical question: could a reductive model with one spatial dimension x and several time dimensions pin down an electron's actual trajectory, bypassing the uncertainty of purely spatial descriptions? The idea came from earlier work in harmonic music theory in the Reaper DAW. Custom templates synchronized frequency-wave patterns to a master BPM clock and tracked them with high-resolution spectroscopy, Blackman–Harris FFT analysis, and a Lorenz attractor used to plot the pulses. These experiments showed "orbital" jumps in interference patterns that stabilized without collapsing, and that stayed stable as long as frequency changes stayed within 100 cents. Translated to quantum mechanics, this suggested that an electron's apparently random motion might reflect temporal interference rather than spatial position. Two time dimensions t_1, t_2 with one spatial dimension x described some behaviour well but lacked the fine-structure constant α, and a third time dimension, t_3, was added to carry it.

The first edition reached hydrogen's ground state with three fitted times. This edition keeps the premise and changes the machinery:

1. It identifies the two clocks the premise needs (Section 3).
2. It supplies the quantization condition the first edition described only in words, and shows that the resulting relation is exact for exactly one force law.
3. It extends the construction from hydrogen to all 118 elements (Section 5).
4. It adds the clocks that end atomic states: spontaneous emission, which runs on a cube of time, and alpha decay, which runs on the Sommerfeld ratio (Section 6).
5. It grades every prediction against literature values, with no parameter adjusted to any atom.
6. It places the result inside the wider Chronowave Temporal Framework and tests two of that framework's own claims directly on the atom: the static temporal volume ($n_c = 0$) and the proposed cubic time-dilation law (Section 7).
7. It finds where cubic time does contribute: in radiation reaction, in orbits weighed in time, and in light from bent paths (Section 8).

The companion interactive instrument, the Chronowave Temporal Engine (Module VIII), runs these calculations live, element by element. The scripts that produced every number are provided with the paper, and every equation below is followed by a boxed worked example that carries a

real input through to a checkable number, so a reader who wants to try a different element, ion or quantum number can follow the same steps (Appendix C lists every constant used).

2 What the first edition established, and what it left open

The first edition's instincts were sound; its machinery was incomplete. Each point below states what the first edition says and the correction this edition makes.

A1. Normalization (1st-ed. eq. (5)–(6)). The wavefunction $\Psi = A\exp[i(\omega_1 t_1 + \omega_2 t_2 + \omega_3 t_3)]$ is a pure phase, so $|\Psi|^2 = A^2$ is constant, and the normalization integral over all of (t_1, t_2, t_3) diverges for every $A \neq 0$. *Fix:* normalize over one temporal cell of volume $V_T = t_1 t_2 t_3$, so that $A = V_T^{-1/2}$. A constant normalization then requires a constant cell volume.

Worked example 1: the normalization constant

Take the first edition's Final-row times, $t_1 = 1.84 \times 10^{-22}$ s, $t_2 = 1.29 \times 10^{-20}$ s and $t_3 = 5.35 \times 10^{-13}$ s. Step 1: $t_1 t_2 = 2.3736 \times 10^{-42}$ s^2. Step 2: $V_T = (2.3736 \times 10^{-42})(5.35 \times 10^{-13}) = 1.2699 \times 10^{-54}$ s^3. Step 3: $A = V_T^{-1/2} = 1/\sqrt{1.2699 \times 10^{-54}} = 8.8740 \times 10^{26}$ s$^{-3/2}$. Check: $A^2 V_T = 1$, so $|\Psi|^2$ integrates to one over the cell. *Try it:* any rescaling of the three times that keeps their product fixed (say $t_1 \to 2t_1$, $t_3 \to t_3/2$) leaves A unchanged; only the product $t_1 t_2 t_3$ matters.

A2. The quantization condition (1st-ed. eq. (27)). $E_n = \hbar\omega_n$, with ω_n "the allowed temporal frequencies that produce constructive interference", names a condition without writing it down. No operator, boundary condition or potential selects the ω_n. *Fix:* Section 3 supplies the Dirac bound-state condition and its temporal reading.

A3. Tables 1, 3 and 4. Tables 1 and 4 are placeholders with every entry zero. In Table 3 the energies $E_1 = h \times 1$ Hz and $E_2 = h \times 1.5$ Hz are the same in every row, so the table shows no dependence on t_1, t_2 or x. The first edition's text describes x as "angular rotation in metres", which mixes an angle with a length. Computed tables replace all three.

A4. The refinement ladder (Table 5). The Initial, Secondary, Tertiary and Final rows adjust t_1, t_2, t_3 against the known energy [FITTED]. The first three rows list the same predicted energy, 2.18×10^{-18} J, while their errors fall from 5174% to 0.0000431%. The Tertiary and Final rows share the error digit for digit although their times differ. The Final row's energy entry, 2.07×10^{16}, equals 2.18×10^{-18} J$/\hbar$ in rad/s. The last step therefore changed the *representation* (energy re-read as a temporal rate), not the fit. Section 9 places these times on the derived clock ladder.

Worked example 2: the Final row's unit change

$\hbar = 1.054571817 \times 10^{-34}$ J s. Dividing the quoted energy by it, 2.18×10^{-18} J$/1.054571817 \times 10^{-34}$ J s $= 2.0672 \times 10^{16}$ rad/s, matches the Final row's entry. *Try it:* dividing instead by $h = 6.626070 \times 10^{-34}$ J s converts the same energy to an ordinary frequency, 3.290×10^{15} Hz – the Rydberg frequency cR_∞ found in A6 below.

A5. The worked error example (1st-ed. eq. (29)). The arithmetic $|2.18 - 2.179|/2.179 = 0.046\%$ is correct, but it disagrees with the 0.0000431% in Table 5. The procedure text also refers to Equation (1) where it means Equation (28). This edition reports signed errors against sourced reference values.

Worked example 3: checking the arithmetic, not the fit

$|2.18 - 2.179|/2.179 = 0.000459 = 0.0459\%$: the arithmetic the first edition performs is correct, but the error printed one column over in Table 5 is a different, far smaller number from a later refinement step. *Try it:* the general form is $|E_{\text{calc}} - E_{\text{ref}}|/E_{\text{ref}}$; substituting this edition's own numbers (Table 1) reproduces the ppm-level errors quoted throughout.

A6. Two findings in Table 5. The Initial row sets $t_1 = 1.29 \times 10^{-21}$ s. That is the electron's Compton time, $\hbar/mc^2 = 1.288 \times 10^{-21}$ s, to the three digits given, as the first edition's text says ("the Compton time of the electron as a baseline"). The first edition began where this one does. Table 5's pair $(\omega_1, \omega_2) = (3.29, 6.58)$ has no units in the first edition. Read in units of 10^{15} Hz, which is our inference, it is the Rydberg frequency $cR_\infty = 3.2898 \times 10^{15}$ Hz and its octave.

Worked example 4: the Compton time and the Rydberg frequency

Compton time. $\hbar = 6.582119569 \times 10^{-16}$ eV s and $mc^2 = 510\,998.95069$ eV give $t_C = \hbar/mc^2 = 1.288089 \times 10^{-21}$ s, which rounds to the Initial row's 1.29×10^{-21} s. *Rydberg frequency.* $R_\infty = 10\,973\,731.568157$ m^{-1} and $c = 299\,792\,458$ m/s give $cR_\infty = 3.289842 \times 10^{15}$ Hz, matching 3.29; its octave $2cR_\infty = 6.579684 \times 10^{15}$ Hz matches 6.58. *Try it:* $h \cdot cR_\infty = (4.135667696 \times 10^{-15}\,\text{eV s})(3.289842 \times 10^{15}\,\text{Hz}) = 13.6057$ eV, the binding of a non-relativistic hydrogen atom with an infinitely heavy nucleus.

A7. Spin sector (1st-ed. eq. (7)–(23)). This is retained in Section 10 with three corrections, to 1st-ed. eq. (9) and (23), to (11)–(13), and to the phase equation of the Lagrangian (20).

3 Methodology: two clocks, one phase

3.1 The premise, and what makes it exact

The electron carries a clock of its own. This is the phase of its rest-energy oscillation, which advances at the Compton frequency $\omega_C = mc^2/\hbar$ in its *proper* time τ. A stationary state carries a second clock, its energy phase, which advances at $E/\hbar$ in *coordinate* time t. Setting the two phases equal gives

$$\underbrace{\omega_C\,\tau}_{\text{proper clock}} \;=\; \underbrace{\frac{E}{\hbar}\,t}_{\text{coordinate clock}} \qquad\Longrightarrow\qquad E \;=\; mc^2\left\langle \frac{d\tau}{dt} \right\rangle. \tag{1}$$

Worked example 5: the premise for hydrogen's ground state

Inputs (CODATA 2022): $\alpha = 7.2973525643 \times 10^{-3}$, $mc^2 = 510\,998.95069$ eV, $\hbar = 6.582119569 \times 10^{-16}$ eV s; infinitely heavy nucleus. *Step 1, the clock rate* (eq. (4) below, with $Z = 1$, $n = 1$): $\langle d\tau/dt \rangle_{1s} = \sqrt{1-\alpha^2} = \sqrt{1 - 5.3251354 \times 10^{-5}} = 0.999973373968$. *Step 2, the energy:* $E = mc^2\langle d\tau/dt\rangle = 510\,998.95069 \times 0.999973373968 = 510\,985.34482$ eV. *Step 3, the two clocks agree:* $\omega_C = mc^2/\hbar = 7.763441 \times 10^{20}$ rad/s runs in proper time; $E/\hbar = 7.763234 \times 10^{20}$ rad/s runs in coordinate time. Their ratio is the clock rate, $7.763234/7.763441 = 0.999973$. *Step 4, the binding:* $B = mc^2 - E = 13.605874$ eV, against 13.605693 eV for Bohr ($\alpha^2 mc^2/2$); the 0.18 meV difference is the relativistic part. *Try it:* for He$^+$ use $Z\alpha$ in place of α: $\sqrt{1-(2\alpha)^2} = 0.999893492$, so $B = mc^2(1 - 0.999893492) = 54.4257$ eV (infinitely heavy nucleus; the reduced mass of Section 4 lowers it to 54.4182 eV).

Equation (1) defines what the energy means in temporal terms [FORCED-BY-DEFINITION]. Whether it holds is a question about dynamics. For the Dirac Hamiltonian $H = c\,\boldsymbol{\alpha}\cdot\mathbf{p} + \beta mc^2 + V(r)$,

scaling $r \to \lambda r$ gives the Dirac virial theorem, first derived by Fock [2], $\langle c\,\boldsymbol{\alpha}\cdot\mathbf{p}\rangle = \langle rV'\rangle$ [1], and therefore

$$E \;=\; mc^2\langle\beta\rangle + \langle V + rV'\rangle \qquad \text{[FORCED-BY-MECHANICS]}. \tag{2}$$

For a power-law potential $V \propto r^k$ the last term equals $(k+1)\langle V\rangle$. It vanishes for every bound state if and only if $k = -1$. So $E = mc^2\langle\beta\rangle$ holds exactly for the inverse-square force and for no other. For a free wave packet $\langle\beta\rangle = 1/\gamma = d\tau/dt$, and in a bound state we *identify* $\langle\beta\rangle$ with the electron's average clock rate, $\langle d\tau/dt\rangle \equiv \langle\beta\rangle$ [FORCED-BY-DEFINITION]. The companion calculations check the dynamical statement by integrating the radial Dirac equations: $\langle\beta\rangle$ equals E/mc^2 to 6×10^{-11} over 45 states with $Z = 1$–92, and for $V \propto r^{-0.9}, \ldots, r^{-0.25}$ the mismatch equals $(k+1)\langle V\rangle$ to every printed digit.

The classical statement. Classically the premise is already a theorem. For any closed system of charges in finite motion, the total energy is the time average of $\sum_a m_a c^2\sqrt{1 - v_a^2/c^2}$: this is Landau and Lifshitz's relativistic virial theorem [3], with the charges' self-energy renormalized as Shafranov showed [4]. For the one-electron atom, Gaite writes it as $E/mc^2 = \langle\sqrt{1 - v^2/c^2}\rangle$ [5]. The square root is $d\tau/dt$, so eq. (1) is that theorem read as a statement about clocks; what the Dirac equation adds is that the reading survives quantization, with $\langle\beta\rangle$ in place of the time average.

Why the identification is more than a reading. The second edition called $\langle d\tau/dt\rangle \equiv \langle\beta\rangle$ "a reading, not a derivation", because proper time has no operator. A firmer statement is available. mc^2 enters the Dirac Hamiltonian only through βmc^2, so the Hellmann–Feynman theorem gives, for any bound state and any potential,

$$\langle\beta\rangle = \frac{\partial E}{\partial(mc^2)} \qquad \text{[FORCED-BY-MECHANICS]}. \tag{3}$$

For the Coulomb field this is a known route to $E = mc^2\langle\beta\rangle$ [6, 7]. Classically, $\partial E/\partial(mc^2)$ at fixed action is exactly the time average of $\sqrt{1 - v^2/c^2}$, the clock rate of the moving particle; Hellmann–Feynman is its quantum counterpart. So $\langle\beta\rangle$ *is* the response of the energy to the rest-energy clock, with no further identification. Equation (1) then reads $E = mc^2\,\partial E/\partial(mc^2)$: the energy is homogeneous of degree one in mc^2. A dimensional argument shows when this happens. The only other scales are $\hbar$, c and the coupling of V; for the Coulomb potential the coupling Ze^2 combines with $\hbar c$ into the pure number $Z\alpha$, so $E = mc^2\,F(Z\alpha, n, j)$ and the homogeneity is automatic. For any other power law the coupling carries a length, E acquires a second energy scale, and eq. (1) fails by exactly $(k+1)\langle V\rangle$.

Worked example 6: Hellmann–Feynman for U^{91+}

Point nucleus. $E(mc^2) = mc^2\sqrt{1 - (Z\alpha)^2}$ is linear in mc^2, so $\partial E/\partial(mc^2) = \sqrt{1 - (92\alpha)^2} = 0.7411346274 = E/mc^2$, which is the clock rate of eq. (4). *Finite nucleus* (uniform sphere, $r_{\mathrm{rms}} = 5.7508$ fm, $R = \sqrt{5/3}\,r_{\mathrm{rms}} = 7.4243$ fm). Now the nuclear radius is a second scale. Solving the radial Dirac equation at $mc^2(1 \pm 10^{-6})$ and differencing gives $\partial E/\partial(mc^2) = 0.7420670756$; the expectation value of β in the same state is 0.7420670757. They agree to 10^{-10}, as eq. (3) requires, but now $E/mc^2 \neq \langle\beta\rangle$: the gap is the nuclear-volume term worked in Example 11. *Try it:* for Fe^{25+}, $Z\alpha = 0.18973$ and $\sqrt{1 - (Z\alpha)^2} = 0.981836$; that is both E/mc^2 and $\langle\beta\rangle$ for a point nucleus.

Worked example 7: a counterexample: the harmonic oscillator

Take $V = \frac{1}{2}Kr^2$ ($k = 2$), in the non-relativistic limit. The ground state is $E = mc^2 + \frac{3}{2}\hbar\omega$ with $\omega = \sqrt{K/m}$, so $\hbar\omega \propto m^{-1/2}$. *Step 1:* $m\,\partial E/\partial m = mc^2 + \frac{3}{2}\hbar\omega \cdot (-\frac{1}{2}) = mc^2 - \frac{3}{4}\hbar\omega$. *Step 2:* the mismatch is $E - m\,\partial E/\partial m = \frac{9}{4}\hbar\omega$. *Step 3:* by the virial theorem $\langle V\rangle = \frac{3}{4}\hbar\omega$, so $(k+1)\langle V\rangle = 3 \cdot \frac{3}{4}\hbar\omega = \frac{9}{4}\hbar\omega$, the same. The premise fails for the oscillator by exactly the amount eq. (2) predicts. *Try it:* for $V \propto r^k$ the mismatch is $(k+1)\langle V\rangle$; set $k = -1$ and it vanishes.

3.2 The quantization, and every level as a clock rate

The bound-state condition of the Dirac–Coulomb problem supplies what 1st-ed. eq. (27) left in words. Every level (n, j) of a one-electron atom has the clock rate

$$\left\langle \frac{d\tau}{dt} \right\rangle_{nj} = \left[1 + \left(\frac{Z\alpha}{n - (j+\frac{1}{2}) + \sqrt{(j+\frac{1}{2})^2 - (Z\alpha)^2}} \right)^2 \right]^{-1/2} \xrightarrow{j=n-\frac{1}{2}} \sqrt{1 - \left(\frac{Z\alpha}{n}\right)^2}. \qquad (4)$$

For circular states ($j = n - \frac{1}{2}$) the level coincides with the relativistic Sommerfeld orbit, whose speed is exactly $\beta = Z\alpha/n$ [11]. The binding energy is the proper time the electron loses, and a spectral line is the difference of two clock rates:

$$B_{nj} = mc^2\big(1 - \langle d\tau/dt\rangle_{nj}\big), \qquad \hbar\omega_{u\to l} = mc^2\big(\langle d\tau/dt\rangle_u - \langle d\tau/dt\rangle_l\big). \qquad (5)$$

Worked example 8: eq. (4) one step at a time

Write the formula as four steps: (i) $Z\alpha$; (ii) the root $\sqrt{(j+\frac{1}{2})^2 - (Z\alpha)^2}$; (iii) the denominator $D = n - (j+\frac{1}{2}) + \text{root}$; (iv) the rate $[1 + (Z\alpha/D)^2]^{-1/2}$. Then $B = mc^2(1 - \text{rate})$ with $mc^2 = 510\,998.95069$ eV.

level	$Z\alpha$	root	D	$(Z\alpha/D)^2$	rate	B (eV)
H $1s_{1/2}$	0.007297353	0.999973374	0.999973374	5.325419×10^{-5}	0.999973373968	13.605874
H $2s_{1/2}$, $2p_{1/2}$	0.007297353	0.999973374	1.999973374	1.331319×10^{-5}	0.999993343470	3.401480
H $2p_{3/2}$	0.007297353	1.999986687	1.999986687	1.331302×10^{-5}	0.999993343559	3.401435
U^{91+} $1s_{1/2}$	0.671356436	0.741134627	0.741134627	0.8205633	0.741134627	132 279.93
U^{91+} $2p_{1/2}$	0.671356436	0.741134627	1.741134627	0.1486763	0.933041968	34 215.48
U^{91+} $2p_{3/2}$	0.671356436	1.883953433	1.883953433	0.1269890	0.941976716	29 649.84

Reading the table. For 1s, $n = j + \frac{1}{2}$, so D is just the root and the rate collapses to $\sqrt{1-(Z\alpha)^2}$ (the right side of eq. (4)). The $2s_{1/2}$ and $2p_{1/2}$ levels share j, so they share every step: Dirac's degeneracy, lifted only by the Lamb shift. *A line* is a difference of rates (eq. (5)): hydrogen's $2p_{3/2}$–$2p_{1/2}$ fine structure is $mc^2(0.999993343558540 - 0.999993343469921) = mc^2 \times 8.861889 \times 10^{-11} = 4.528416 \times 10^{-5}$ eV $= 10.9497$ GHz; times the reduced-mass factor $\mu/m = 0.9994556794$ it is 4.525951×10^{-5} eV $= 10.9437$ GHz. (Keep 15 digits in the rates here: the line is a difference of two numbers that agree to ten.) *Try it:* C^{5+} 1s has $Z\alpha = 0.0437841$, rate $\sqrt{1 - 0.0437841^2} = 0.999041$, $B = 490.0$ eV.

Worked example 9: proper time lost per day

Seconds lost per day $= (1 - \text{rate}) \times 86\,400$ s. H 1s: $2.662603 \times 10^{-5} \times 86\,400 = 2.3005$ s. U^{91+} 1s: $0.2588654 \times 86\,400 = 22\,366$ s $= 6.2128$ h. Og^{117+} 1s (point nucleus, $Z\alpha = 0.86109$): rate 0.508457, so 42 469 s $= 11.797$ h. *Where the clock stops.* The 1s rate $\sqrt{1-(Z\alpha)^2}$ reaches zero at $Z\alpha = 1$, i.e. $Z = 1/\alpha = 137.036$; the non-relativistic clock $1 - (Z\alpha)^2/2$ reaches zero only at $Z = \sqrt{2}/\alpha = 193.80$. *Try it:* gold, $Z = 79$: $Z\alpha = 0.576491$, rate $= \sqrt{1 - 0.332342} = 0.8171$; the 1s electron of a gold atom ages at 82% of the laboratory rate, losing 4.4 h per day.

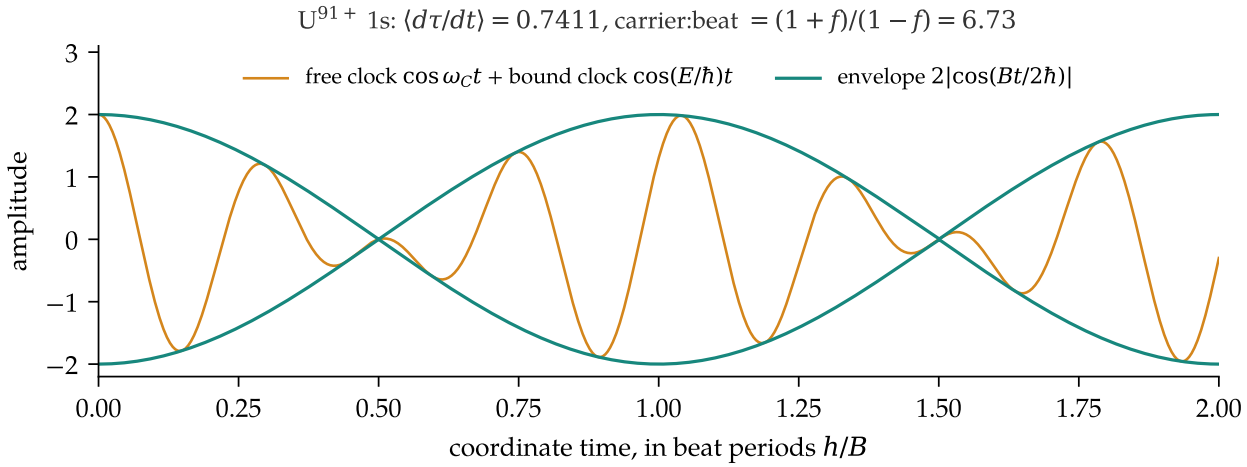


Figure 1: The first edition's Figure 1 rebuilt with physical frequencies: a free electron's clock and a bound state's clock, superposed as in its eq. (26). The envelope beats at B/h. Shown for U^{91+}, where the carrier-to-beat ratio is small enough to draw.

Hydrogen's ground-state electron loses 2.30 s of proper time per day of coordinate time; in U^{91+} it loses 6.2 hours. For a point nucleus the 1s clock stops at $Z\alpha = 1$, that is $Z = 137.04$. A finite nucleus keeps the level bound beyond that. In our uniform-sphere model the energy passes zero near $Z \approx 147$ and $\langle\beta\rangle$ near $Z \approx 153$; the two differ because $E = mc^2\langle\beta\rangle$ fails inside a finite nucleus. The level reaches the negative-energy continuum near $Z \approx 170$ [12].

3.3 The first edition's interference, with physical frequencies

The first edition's eq. (26), $\psi = 2A\cos\left(\frac{\omega_1+\omega_2}{2}t\right)\cos\left(\frac{\omega_1-\omega_2}{2}t\right)e^{ikx}$, becomes physical when ω_1 is a free electron's clock ω_C and ω_2 is the bound state's clock $E/\hbar$, both read on the laboratory clock. By eq. (1) the bound electron's own proper clock never beats against its energy phase. What beats is the bound clock against the free one. The envelope then oscillates at B/h, so the binding energy is the beat between a free and a bound electron. For hydrogen this beat is the Lyman limit, the photon that ionizes the atom. The carrier-to-envelope ratio is $(1+f)/(1-f)$ with $f = \langle d\tau/dt\rangle$: about 75,000 for hydrogen and 6.73 for U^{91+} (Figure 1).

Worked example 10: the beat between a free and a bound electron

The sum of $\cos\omega_1 t$ and $\cos\omega_2 t$ is a carrier at $(\omega_1+\omega_2)/2$ under an envelope at $(\omega_1-\omega_2)/2$; with $\omega_1 = mc^2/\hbar$ and $\omega_2 = E/\hbar = f\,mc^2/\hbar$, the envelope's intensity repeats at $(\omega_1-\omega_2)/2\pi = B/h$. *Hydrogen* (reduced mass, $B = 13.598468$ eV): $B/h = 13.598468/4.135667696 \times 10^{-15} = 3.288095 \times 10^{15}$ Hz, and the corresponding wavelength $hc/B = 1239.841984/13.598468 = 91.1751$ nm is the Lyman limit. The carrier-to-envelope ratio is $(1+f)/(1-f)$ with $f = 0.999973374$: $1.999973374/2.6626 \times 10^{-5} = 75\,113$. *$U^{91+}$:* $f = 0.741134627$, so $(1+f)/(1-f) = 1.741134627/0.258865373 = 6.726$; seven carrier cycles per beat, few enough to draw (Figure 1). *Try it:* for He^+ ($f = 0.99989349$), the ratio is $1.99989349/1.0651 \times 10^{-4} \approx 18\,777$, a quarter of hydrogen's, because $1-f$ grows as Z^2.

3.4 Many electrons

The whole atom. Consider any mean-field description in which every interaction is homogeneous of degree -1 in the coordinates. Dirac–Hartree–Fock and exchange-only Dirac–Slater theory are examples, and the frequency-independent Breit interaction is also of degree -1. In such a model, scaling all coordinates together gives $\langle \sum_i c\,\boldsymbol{\alpha}_i \cdot \mathbf{p}_i \rangle = -\langle V \rangle$, Bahcall's virial theorem for many-electron Dirac systems [8], and the premise holds for the whole atom:

$$E_{\text{atom}} \;=\; \sum_{i=1}^{Z} mc^2 \Big(\langle d\tau/dt \rangle_i - 1 \Big) \;+\; E_{\text{inside nucleus}} \tag{6}$$

[FORCED-BY-MECHANICS, WITHIN THAT CLASS]. Inside the class, the one departure from inverse-square forces is the interior of the nucleus. There the charge is spread out and the force grows linearly, so this term is carried separately. It is 7×10^{-10} of the total energy for hydrogen, 6.5×10^{-6} for iron, 9×10^{-4} for uranium and 6×10^{-3} for oganesson. The identity does *not* hold for density functionals with correlation (which are not homogeneous), for no-pair projected many-body Hamiltonians, or for hyperfine and QED potentials.

The nuclear term has a simple closed form. A uniformly charged sphere of radius R introduces one length scale beyond $\hbar/mc$, and the same scaling argument that gave eq. (2) now gives

$$E_{\text{inside nucleus}} = \langle V + rV' \rangle = -R\,\frac{\partial E}{\partial R} \qquad \text{[FORCED-BY-MECHANICS]}: \tag{7}$$

the nuclear-volume term is the energy's response to the size of the nucleus. This is the relativistic counterpart of Slater's molecular virial theorem, with the nuclear radius in place of the internuclear distances [34]. Matsuoka and Koga derived this finite-nucleus correction to the relativistic virial theorem of atoms for four nuclear models [9]; eq. (7) is its Hellmann–Feynman form for a charge distribution with a single radius.

Worked example 11: the nuclear-volume term in U^{91+}

Inputs: $Z = 92$, uniform sphere with $r_{\text{rms}} = 0.836A^{1/3} + 0.570 = 5.7508$ fm ($A = 238$), so $R = \sqrt{5/3} \times 5.7508 = 7.4243$ fm. *Step 1, the shift:* the point-nucleus 1s energy is $E - mc^2 = -132\,279.934$ eV; with the sphere it is $-132\,086.144$ eV, so the finite size lifts the level by 193.790 eV. *Step 2, three ways to the same term:* $E - mc^2\langle\beta\rangle = -282.6231$ eV; $\langle V + rV' \rangle = -282.6205$ eV (zero outside the nucleus, so only the interior contributes); $-R\,\partial E/\partial R = -282.6234$ eV (from solving at $R(1 \pm 10^{-4})$). All three agree to 3 meV. *Step 3, a check by scaling:* near the origin the 1s density goes as $r^{2\gamma-2}$ with $\gamma = \sqrt{1-(Z\alpha)^2} = 0.741135$, so the shift grows as $R^{2\gamma}$ and $R\,\partial(\text{shift})/\partial R \approx 2\gamma \times 193.790 = 287.2$ eV, within 2% of Step 2. *Try it:* the term is 9×10^{-4} of the whole uranium atom's energy and 7×10^{-10} of hydrogen's, because the 1s density at the nucleus grows as Z^3 and the nucleus itself grows as $A^{1/3}$.

Each electron: clock, shelter and exchange. For one electron in the atom's mean field, eq. (2) reads

$$\varepsilon_i = \underbrace{mc^2\big(\langle d\tau/dt \rangle_i - 1\big)}_{\text{clock}} + \underbrace{\big\langle U_{\text{out}} \big\rangle_i}_{\text{shelter}} + \underbrace{\big\langle \partial_r (rV_{xc}) \big\rangle_i}_{\text{xc}}. \tag{8}$$

Here $U_{\text{out}}(r) = \partial_r [rV_{\text{es}}(r)]$ is exactly the electrostatic potential energy due to all charge lying farther from the nucleus than r. That charge is the outer electrons and, for the part of an orbital inside the nucleus, the nuclear charge beyond r. By the shell theorem this charge exerts no net force at r. Semiclassically, then, it does not change the electron's local speed, and so it does not

change its clock; it only lifts the energy. The clock is set by the charge inside, the shelter by the charge outside. The shelter also includes the electron's own cloud beyond r. Exact exchange would remove that self part; the local-density approximation removes only part of it. For hydrogen the model removes 2.7 of 8.1 eV, which is the familiar self-interaction error, and it is why the model's hydrogen 1s clock reads 2.15 s/day against the exact 2.30. Equation (8) is the one-electron virial theorem in the mean field. Froese Fischer and Godefroid give the corresponding orbital relations in Dirac–Hartree–Fock theory, where a potential $-Z(r)/r$ contributes a correction $\langle W\rangle = \langle Z'(r)\rangle$ [10]; shelter and exchange together are $-\langle W\rangle$, and the ledger splits that term by where the charge lies.

Worked example 12: the clock–shelter ledger of helium

Helium is the smallest atom with shelter, and in Hartree–Fock (exact exchange, non-relativistic) the self part cancels exactly, so every term can be computed by hand to good accuracy. *Hand version* (both electrons in a hydrogenic 1s orbital with $Z_{\text{eff}} = Z - \frac{5}{16} = \frac{27}{16}$; 1 hartree = 27.211386 eV): (i) total energy $E = -Z_{\text{eff}}^2 = -2.8477$ hartree $= -77.489$ eV; (ii) kinetic energy per electron $\langle T\rangle = Z_{\text{eff}}^2/2 = 1.4238$ hartree $= 38.744$ eV, so the clock term is -38.744 eV, and per day each electron loses $\langle T\rangle/mc^2 \times 86\,400 = 6.551$ s; (iii) the repulsion between the two clouds is $J = \frac{5}{8}Z_{\text{eff}} = 1.0547$ hartree; for one electron the shelter from the other is $J/2 = \frac{5}{16}Z_{\text{eff}} = 0.5273$ hartree $= 14.350$ eV; (iv) clock + shelter $= -38.744 + 14.350 = -24.39$ eV. The ledger is exact only for self-consistent orbitals, so the hand value is an estimate; the self-consistent one follows. *Self-consistent version* (numerical Hartree–Fock, `we_helium.py`): $E = -2.8616799$ hartree (literature -2.8616800), $\langle T\rangle = 1.4308392$ hartree $= 38.935$ eV per electron, $J = 1.0257688$ hartree, shelter $J/2 = 13.956$ eV. *The whole-atom ledger* (eq. (6); non-relativistic, point nucleus, so $E_{\text{inside}} = 0$): the two clock terms sum to $2 \times (-38.935) = -77.870$ eV, which is the total energy -77.8703 eV, to 4×10^{-5} eV. *The orbital ledger:* clock + shelter $= -38.935 + 13.956 = -24.979$ eV, which is the orbital energy $\varepsilon_{1s} = -0.9179555$ hartree $= -24.979$ eV (to 2×10^{-5} eV). The shelter cancels 35.8% of the clock. By Koopmans' theorem the ionization energy is $-\varepsilon = 24.98$ eV; the measured value is 24.587 eV. *Try it:* the hand version for Li^+ ($Z = 3$, $Z_{\text{eff}} = 2.6875$): clock -98.27 eV, shelter $J/2 = 22.85$ eV, estimate $\varepsilon \approx -75.4$ eV (Li^+'s measured ionization energy is 75.64 eV).

The computation. Each element $Z = 1$–118 was computed as a self-consistent relativistic atom:

- **Physics:** the radial Dirac equation for every electron, in a uniform-sphere nucleus (radii in Appendix A).
- **Mean field:** the Hartree field of the electrons' own density plus spin-polarized local-density exchange and correlation (Kohn–Sham [13], Perdew–Zunger [14]) [ASSUMED].
- **Inputs:** Z and the NIST ground configuration [MEASURED]; beyond $Z = 103$ the configuration is a prediction [ASSUMED]. Nothing is fitted to any atom.
- **Subshells:** each shell is split into its two j subshells by statistical weight, except that the 6p and 7p shells are filled $j = \frac{1}{2}$ first (jj coupling) [ASSUMED].
- **Ionization and K-shell binding:** Slater's transition state [15], the orbital energy at half occupation (Janak's theorem [16]).
- **The departing electron:** one electron is removed from each of the three least-bound shells in turn, and the smallest ionization energy is kept [ASSUMED]. Ion ground states that require rearranging the remaining electrons (V^+ $3d^4$, Co^+ $3d^8$, Ni^+ $3d^9$, La^+ $5d^2$, Ce^+ $4f5d^2$) lie outside this rule.

The ledger (6) is checked in the exchange-only (Dirac–Slater) version of the same atoms, where every interaction is homogeneous and the theorem holds identically.

3.5 Decay clocks

Radiative decay: a cube of time. The electric-dipole emission rate $A = 4\alpha\omega^3|\mathbf{r}_{fi}|^2/3c^2$, rewritten in time and nothing else, is

$$\tau_{E1} = \frac{1}{A} = \frac{3}{32\pi^3\alpha}\,\frac{T^3}{t_d^2}, \qquad t_d = \frac{|\mathbf{r}_{fi}|}{c} \tag{9}$$

[FORCED-BY-MECHANICS]. Here T is the period of the emitted light and t_d is the light-crossing time of the transition dipole, with $|\mathbf{r}_{fi}|^2$ summed over final sublevels. For a p→s line, $|\mathbf{r}_{fi}|^2 = R^2/3$, where R is the radial integral. The lifetime is thus the cube of the period divided by the square of the dipole time, scaled by $1/\alpha$. For one-electron ions $T \propto Z^{-2}$ and $t_d \propto Z^{-1}$, so $\tau \propto Z^{-6}/Z^{-2} = Z^{-4}$, and the excited states of heavy one-electron ions die fastest.

Worked example 13: the cube law for hydrogen's 2p state

Step 1, the period. The Lyman-α wavelength is $\lambda = 121.567$ nm, so $T = \lambda/c = 121.567 \times 10^{-9}/299\,792\,458 = 4.05504 \times 10^{-16}$ s. *Step 2, the dipole time.* The 2p→1s radial integral is $R = 128\sqrt{6}/243 = 1.290266$ bohr. With $a_0 = 5.29177210544 \times 10^{-11}$ m, $|\mathbf{r}_{fi}| = R\,a_0/\sqrt{3}$ and $t_d = |\mathbf{r}_{fi}|/c = 1.290266 \times 5.29177 \times 10^{-11}/(1.732051 \times 299\,792\,458) = 1.31492 \times 10^{-19}$ s. *Step 3, the powers.* $T^3 = 6.66784 \times 10^{-47}$ s^3; $t_d^2 = 1.72901 \times 10^{-38}$ s^2; $32\pi^3\alpha = 32 \times 31.00628 \times 0.0072973526 = 7.24044$. *Step 4, the lifetime.* $\tau = 3T^3/(32\pi^3\alpha\, t_d^2) = 3 \times 6.66784 \times 10^{-47}/(7.24044 \times 1.72901 \times 10^{-38}) = 1.5979 \times 10^{-9}$ s. The NIST A-value 6.2649×10^8 s^{-1} gives $1/A = 1.5962$ ns: +0.1%. *Try it (scaling):* for He$^+$ 2p, $T \to T/Z^2$ and $t_d \to t_d/Z$, so $\tau \to \tau/Z^4 = 1.5979/16$ ns $= 99.87$ ps. *Try it (a real atom):* Na 3p$_{3/2}$, with $R = 4.295$ bohr from the computed atom and the measured period $T = 1.965$ fs: $t_d = 4.295 \times 5.29177 \times 10^{-11}/(\sqrt{3}\,c) = 0.4377$ as and $\tau = 3(1.965 \times 10^{-15})^3/(7.24044 \times (4.377 \times 10^{-19})^2) = 16.41$ ns, against 16.23 ns measured (Table 2 prints 16.42 from unrounded inputs).

Alpha decay: the Sommerfeld count. An alpha particle (charge $Z_2 = 2$) with asymptotic speed βc escapes a daughter nucleus of charge Z_1 and mass number A_d through the Coulomb barrier. It does so after e^{2G} ticks of a nuclear clock $t_a = 2R/\beta c$ [17, 18]:

$$t_{1/2} = \ln 2\; t_a\, e^{2G}, \qquad 2G = 4\eta\left[\arccos\sqrt{x} - \sqrt{x(1-x)}\right] \xrightarrow{R\to 0} 2\pi\eta, \qquad \eta = \frac{Z_1 Z_2\,\alpha}{\beta}. \tag{10}$$

Here $x = R/b$, and $b = Z_1 Z_2 e^2/(4\pi\varepsilon_0 Q)$ is the classical turning radius. The Sommerfeld number η is the Coulomb speed divided by the particle's speed; for a bound electron on a circular Sommerfeld orbit it equals n. We take the textbook radius $R = 1.2\,(A_d^{1/3} + 4^{1/3})$ fm and the assault time at the asymptotic speed [ASSUMED].

Worked example 14: the Gamow half-life of U-238, step by step

Inputs. Daughter Th-234: $Z_1 = 90$, $A_d = 234$; alpha: $Z_2 = 2$. Decay energy from AME2020 masses, $Q = 4.2699$ MeV (see the erratum in Appendix F). Atomic masses $M(^4\mathrm{He}) = 4.002603$ u, $M(^{234}\mathrm{Th}) = 234.043600$ u, 1 u $= 931.494$ MeV; $e^2/4\pi\varepsilon_0 = 1.439965$ MeV fm.

1. Reduced mass: $\mu c^2 = \dfrac{4.002603 \times 234.043600}{238.046203} \times 931.494 = 3665.71$ MeV.
2. Speed: $\beta = \sqrt{2Q/\mu c^2} = \sqrt{8.5397/3665.71} = 0.048266$.

3. Sommerfeld number: $\eta = Z_1 Z_2 \alpha/\beta = 180 \times 0.0072974/0.048266 = 27.2142$ (so $2\pi\eta = 170.99$).
4. Radius: $R = 1.2\,(234^{1/3} + 4^{1/3}) = 1.2\,(6.16224 + 1.58740) = 9.29957$ fm.
5. Turning radius: $b = Z_1 Z_2 \times 1.439965/Q = 180 \times 1.439965/4.2699 = 60.7031$ fm; $x = R/b = 0.153198$.
6. The bracket: $\arccos\sqrt{x} = \arccos 0.391405 = 1.168639$; $\sqrt{x(1-x)} = 0.360178$; difference 0.808461.
7. Gamow exponent: $2G = 4\eta \times 0.808461 = 4 \times 27.2142 \times 0.808461 = 88.0064$; $e^{2G} = 1.6622 \times 10^{38}$.
8. Nuclear clock: $t_a = 2R/\beta c = 2 \times 9.29957 \times 10^{-15}/(0.048266 \times 2.99792 \times 10^8) = 1.28537 \times 10^{-21}$ s.
9. Half-life: $t_{1/2} = \ln 2 \times 1.28537 \times 10^{-21} \times 1.6622 \times 10^{38} = 1.481 \times 10^{17}$ s $= 4.693 \times 10^9$ yr.

Measured: 4.463×10^9 yr, so $\log_{10}(4.693/4.463) = +0.02$ decades. Note how much the exponent dominates: the alpha particle strikes the barrier $e^{2G} \approx 10^{38.2}$ times before escaping, each strike lasting about one Compton time of the electron. *Try it:* Po-212 (daughter Pb-208, $Z_1 = 82$, $Q = 8.9542$ MeV): $\mu c^2 = 3658.00$ MeV, $\beta = 0.069969$, $\eta = 17.1042$, $R = 9.01487$ fm, $b = 26.3736$ fm, $x = 0.341814$, bracket 0.472031, $2G = 32.2949$, $t_a = 8.595 \times 10^{-22}$ s, $t_{1/2} = 6.32 \times 10^{-8}$ s against 2.94×10^{-7} s measured (-0.67 decades). Roughly doubling Q (with Z_1 falling from 90 to 82) lowers η by 37% and the half-life by 24 orders of magnitude.

3.6 Error metric

The first edition's eq. (28) is kept, but errors are reported signed. Grading targets are literature values, and none is used as an input. Some are measurements: the deuterium line, the X-ray lines, the K-shell binding energies, the alkali lifetimes and the half-lives. Hydrogen's 2p lifetime is a theoretical reference value. Others are NIST reference values: the ionization energies, which for H and He^+ come from QED theory anchored to spectroscopy.

Worked example 15: a signed error

Signed error $=$ (prediction $-$ reference)/reference. For hydrogen's ionization energy, the proper-time atom gives 13.598468301 eV and NIST gives 13.598434599702 eV: $(13.598468301 - 13.598434599702)/13.598434599702 = 3.37013 \times 10^{-5}/13.5984346 = +2.478 \times 10^{-6} = +2.478$ ppm. The sign says the prediction is too deep, which is the direction the Lamb shift corrects. *Try it:* Bohr's 13.598287264 eV gives -10.835 ppm; the sign flips because the non-relativistic atom is too shallow.

4 Results: one-electron atoms

4.1 Hydrogen, deuterium, He^+ and U^{91+}

Table 1 and Figure 2 compare the non-relativistic (Bohr) atom with the proper-time atom of eq. (4), using reduced masses and CODATA constants. In every case the proper-time atom leaves only the Lamb shift: radiative QED, which is beyond any clock-rate construction, plus the finite nuclear size for heavy ions. The Lamb shifts used are the measured 1S shifts of hydrogen (8172.9 MHz) and deuterium (8183.8 MHz) [19]. For He^+ the hydrogen value is scaled by the leading-order QED law $Z^4[\ln(Z\alpha)^{-2} - \ln k_0 + \frac{19}{30}]$, with the 1S Bethe logarithm $\ln k_0 = 2.984$.

Worked example 16: the hydrogen row of Table 1, column by column

Reduced mass. $m_p/m_e = 1836.152673426$, so $\mu/m = 1/(1 + 1/1836.152673426) = 0.9994556794$. *Bohr column.* $B = (\mu/m)\,R_\infty hc = 0.9994556794 \times 13.605693123 = 13.598287264$ eV; error $(13.598287264 - 13.598434600)/13.598434600 = -10.835$ ppm. *Proper-time column.* $B = (\mu/m)\,mc^2(1 - \sqrt{1-\alpha^2}) =$

Table 1: One-electron atoms with zero free parameters. Errors are prediction minus reference.

target (reference)	Bohr	proper time	Lamb shift	after it
H ionization, 13.5984346 eV (NIST)	−10.83 ppm	+2.48 ppm	2.49 ppm	−0.01 ppm
He^+ ionization, 54.4177655 eV (NIST)	−45.08 ppm	+8.18 ppm	8.10 ppm	+0.08 ppm
D Lyα 1S–$2P_{3/2}$, 121.53375 nm [20]	+13.38 ppm	−3.26 ppm	−3.32 ppm	+0.05 ppm
U^{91+} 1s binding [22]	−12.64%	+0.35%	0.35%	by definition

$0.9994556794 \times 13.605874258 = 13.598468301$ eV; error +2.478 ppm (the signed-error example in Section 3). *Lamb-shift column.* The measured 1S Lamb shift is 8172.874 MHz; times h, $8172.874 \times 10^6 \times 4.135667696 \times 10^{-15} = 3.380029 \times 10^{-5}$ eV, which is 2.486 ppm of the binding. *After it.* QED raises the 1S level, so it reduces the binding: $13.598468301 - 0.000033800 = 13.598434501$ eV, error −0.007 ppm. *Try it:* redo the row with $\mu/m = 1$ (an infinitely heavy proton); the Bohr error becomes +534 ppm, because the reduced mass is a far larger effect than relativity in hydrogen.

Worked example 17: scaling the Lamb shift from H to He^+

The leading-order QED law is $\Delta E_{\text{Lamb}} \propto Z^4 [\ln(Z\alpha)^{-2} - \ln k_0 + \frac{19}{30}]$, with the 1S Bethe logarithm $\ln k_0 = 2.984128$. H: $\ln(\alpha^{-2}) = 9.840487$, bracket $= 9.840487 - 2.984128 + 0.633333 = 7.489693$. He^+: $\ln((2\alpha)^{-2}) = 9.840487 - \ln 4 = 8.454193$, bracket $= 6.103398$. Scale factor $= 2^4 \times 6.103398/7.489693 = 13.03850$, so the He^+ shift is $13.03850 \times 3.380029 \times 10^{-5} = 4.40705 \times 10^{-4}$ eV $= 8.099$ ppm of 54.4177655 eV. The proper-time error +8.178 ppm minus 8.099 ppm leaves +0.079 ppm. *Why the residual ratio is 4.17 and not 4:* the Dirac parts are −13.313 and −53.255 ppm (ratio 4.0002), but the Lamb shift cancels 18.7% of the Dirac part in H and 15.2% in He^+, since its bracket shrinks with Z.

Herzberg's deuterium line. Herzberg [20] measured the unresolved deuterium Lyman-α doublet at 1215.3378 ± 0.0003 Å and reported a ground-state Lamb shift of 0.262 ± 0.038 cm^{-1}. A −0.0003 Å profile correction gives 1215.3375 Å for the 1S–$2P_{3/2}$ component, as reconstructed in [21]. Equation (4), with the reduced mass, gives 82 281.939 cm^{-1} for this transition, the Dirac value of that analysis. The 0.268 cm^{-1} gap to Herzberg's line is his ground-state Lamb shift. Adding the measured 1S Lamb shift [19] closes it to +0.05 ppm, within his ±0.0003 Å (±0.25 ppm).

Worked example 18: Herzberg's deuterium line

Reduced mass: $m_d/m_e = 3670.482967655$, $\mu/m = 0.9997276305$. *The line* is a difference of two rates (eq. (5)): $2p_{3/2}$ rate 0.999993343558540, 1s rate 0.999973373968303, difference $1.996959024 \times 10^{-5}$, so $\hbar\omega = (\mu/m)\, mc^2 \times 1.996959024 \times 10^{-5} = 10.201660278$ eV. (As with the fine structure, keep 15 digits in the rates.) In wavenumbers, divide by $hc = 1.239841984 \times 10^{-4}$ eV cm: 82 281.9392 cm^{-1}, i.e. $\lambda =$ 121.533353 nm. *The reference:* $\lambda = 121.53375$ nm is 82 281.6707 cm^{-1}; the gap is 0.2685 cm^{-1}. *The Lamb shift:* 8183.807 MHz $\times h = 3.384551 \times 10^{-5}$ eV $= 0.27298$ cm^{-1}. It lifts the 1S level and so shortens the line: $82\,281.9392 - 0.2730 = 82\,281.6662$ cm^{-1}, $\lambda = 121.533757$ nm, +0.054 ppm from Herzberg's value. *Try it:* with Bohr's line, $\frac{3}{4}(\mu/m)R_\infty hc$, the wavelength is 121.535376 nm, +13.4 ppm: relativity alone is worth 16.6 ppm on this line.

U^{91+}. U^{91+} sets a scale rather than a test: its measured quantity is the Lamb shift itself, 460.2 ± 4.6 eV [22], defined against this very Dirac value. Bohr misses 12.6% of the binding of the heaviest one-electron atom measured. The proper-time atom (point nucleus) misses 0.35%, which is the whole measured Lamb shift. Of that, about 0.15% is the finite size of the nucleus and about 0.2% is QED.

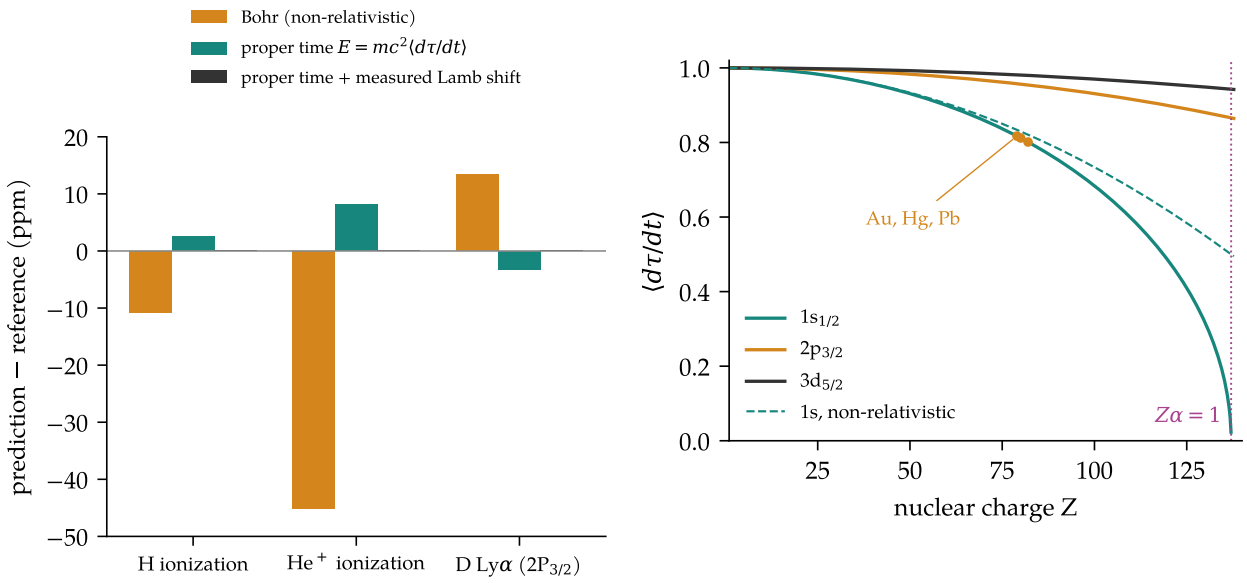


Figure 2: Left: one-electron predictions against reference values; the proper-time atom leaves only the Lamb shift. Right: $\langle d\tau/dt\rangle$ across the table for a point nucleus. The Dirac 1s clock stops at $Z\alpha = 1$. The non-relativistic curve does not stop within the table (it would reach zero only at $Z \approx 194$). Markers: gold, mercury and lead, whose 1s clocks run at 80%–82% speed.

Worked example 19: U^{91+}, where relativity is not a correction

$Z\alpha = 92 \times 0.0072973526 = 0.671356$; $\mu/m = 0.999997695$. *Bohr:* $(\mu/m)(Z\alpha)^2 mc^2/2 = 0.999997695 \times 0.450719 \times 255\,499.48 = 115\,158.32$ eV. *Proper time:* $(\mu/m)\,mc^2(1 - 0.741134627) = 132\,279.63$ eV. *Reference:* the measured Lamb shift 460.2 ± 4.6 eV is defined against this Dirac value, so the reference binding is $132\,279.63 - 460.2 = 131\,819.43$ eV. *Errors:* Bohr −12.639%; proper time +0.349%. Of the 460 eV, 193.8 eV is the finite nucleus (Worked example 11) and the rest is QED; theory gives 463.8 eV in total [23]. *Try it:* the same steps for Au^{78+} ($Z = 79$) give a Bohr binding of 84 913 eV and a Dirac binding of 93 460 eV (infinitely heavy nucleus): relativity adds 10.1%.

The earlier residuals. The non-relativistic residuals are now fully explained. Their Dirac part is −13.31 ppm in H and −53.25 ppm in He^+, a ratio of $4.000 = Z^2$. The Lamb shift grows as $Z^4 \ln(Z\alpha)^{-2}$, a little more slowly than the Dirac term's Z^4, so it cancels 19% of the Dirac term in H but only 15% in He^+. Dirac plus Lamb therefore predicts the He^+/H residual ratio as 4.17; the reference values give 4.16.

4.2 X-ray lines of real atoms

Moseley's law [25], read as a difference of clock rates, is $K\alpha_1 = mc^2[\langle d\tau/dt\rangle_{2p_{3/2}} - \langle d\tau/dt\rangle_{1s}]$ at $Z - 1$, with Moseley's screening $\sigma = 1$ [ASSUMED]. Against 32 $K\alpha_1$ lines from Al to U [26], the RMS error falls from 5.71% (Bohr/Moseley) to 1.05% (Figure 3a). The 32 elements plotted are Al, Si, Ca, Ti, Cr–Zn, Ge, Zr, Mo, Rh–Cd, Sn, I, Ba, Nd, Gd, Yb, Ta, W, Pt–Hg, Pb, Bi and U. Over all 80 $K\alpha_1$ lines from Al to U in the same table the figures are 6.34% and 1.09%, with residuals from −0.92% to +1.53%, so the subset is representative. Moseley's straight line bends away from the data as Z grows, by 14% at uranium, and that bend is the proper-time deficit the non-relativistic law leaves out. The remainder, −0.9% to +1.5% and mostly +0.4% to +1.5%, reflects outer-electron screening.

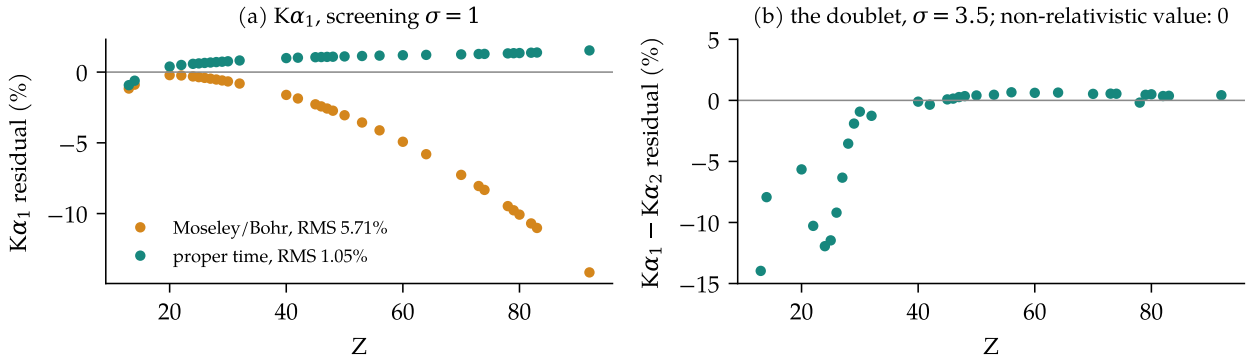


Figure 3: (a) Kα_1 residuals, Bohr/Moseley against proper time. (b) The Kα doublet, which is zero in any non-relativistic atom. Line energies from the X-ray Data Booklet, Table 1-2.

The Kα doublet is a purely relativistic effect: the non-relativistic atom predicts a splitting of exactly zero at every Z. With Sommerfeld's doublet screening $\sigma = 3.5$ [ASSUMED], it comes out within 0.66% for $Z \geq 40$, over splittings from 84 to 3774 eV (Figure 3b). Below $Z \approx 30$ a single screening constant fails (-14% at Al).

Worked example 20: molybdenum's Kα lines

Measured (X-ray Data Booklet, Table 1-2): $K\alpha_1 = 17\,479.34$ eV, $K\alpha_2 = 17\,374.3$ eV, doublet 105.04 eV. Mo has $Z = 42$, so with $\sigma = 1$ the effective charge is $Z - 1 = 41$. *Moseley (Bohr):* $\frac{3}{4}R_\infty hc\,(Z-1)^2 = 0.75 \times 13.605693 \times 1681 = 17\,153.38$ eV, error -1.86%. *Proper time:* $Z_{\text{eff}}\alpha = 41 \times 0.0072973526 = 0.299191$. $2p_{3/2}$ is circular, so its rate is $\sqrt{1-(0.299191/2)^2} = 0.988747247$; 1s rate $\sqrt{1-0.299191^2} = 0.954193101$. $K\alpha_1 = mc^2(0.988747247 - 0.954193101) = 510\,998.95 \times 0.034554146 = 17\,657.13$ eV, error $+1.02\%$. *The doublet*, with $\sigma = 3.5$: $Z_{\text{eff}} = 38.5$, $Z_{\text{eff}}\alpha = 0.280948$. $2p_{1/2}$: root $\sqrt{1-0.280948^2} = 0.959722970$, $D = 1 + 0.959722970 = 1.959722970$, $(0.280948/1.959722970)^2 = 0.020552410$, rate $= 1/\sqrt{1.020552410} = 0.989879531$. $2p_{3/2}$: rate $\sqrt{1-(0.280948/2)^2} = 0.990084363$. Doublet $= mc^2(0.990084363 - 0.989879531) = 104.67$ eV, error -0.35%. A non-relativistic atom gives exactly zero. *Try it:* silver ($Z = 47$, measured $K\alpha_1 = 22\,162.92$ eV): $Z_{\text{eff}}\alpha = 46\alpha = 0.335678$, and the same two square roots give 22 401 eV, $+1.07\%$.

5 Results: all 118 elements

5.1 The whole-atom ledger

For every element from hydrogen to oganesson, the exchange-only Dirac–Slater atom satisfies eq. (6) to 1.1×10^{-7} or better, with a nuclear-volume term of at most 0.6% (oganesson). The total energy of each of these atoms is the sum of its electrons' proper-time deficits plus that term. The closure checks an identity of the model; it is not a measure of the model's accuracy. With correlation included, the identity picks up a residual from the correlation functional, which is not homogeneous: 2.5% of the total for hydrogen, 0.5% for nitrogen, and about 3×10^{-4} by gold.

5.2 Ionization and K-shell binding

Figure 4 and Appendix E compare the computed first ionization energies with the reference values. Over the 99 elements with a NIST reference value that is not merely an estimate, the RMS error is

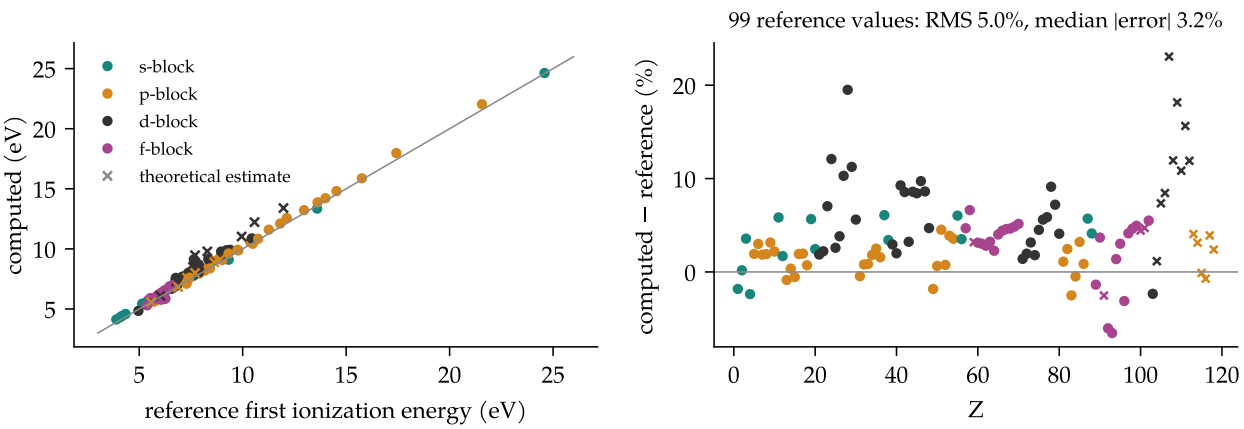


Figure 4: First ionization energies for all 118 elements, computed with no per-atom parameters, against NIST reference values ($Z \leq 108$) and theoretical estimates (crosses; $Z \geq 109$ and the NIST estimates).

5.0% and the median absolute error is 3.2%. For hydrogen the NIST value is QED theory anchored to spectroscopy. The rule for the departing electron gives the known departing shell where one exists. Lutetium, hafnium, yttrium and palladium lose a d electron (Lu^+ is $4f^{14}6s^2$), and actinium, protactinium, uranium, neptunium and curium lose a 6d. The N/O inversion of the half-filled shell is reproduced; the P/S and As/Se inversions are not.

The largest misses, 8–20%, are transition metals: Ni, Cr, Cu, Co, Pd, Nb, Pt, Ag, Ru, Mo and Rh. Local-density exchange is known to misplace the balance between d and s electrons. For nickel most of the +19.5% has a second cause: its true ion ground state ($3d^9$) requires rearranging the remaining electrons, which the one-electron removal cannot reach. For $Z = 104$–118 no measurement exists, and the comparison is with theoretical estimates (marked † in Appendix E).

K-shell binding across the 83 elements from neon to uranium [26] agrees to 0.46% RMS, with a mean of +0.42% (Figure 5, left). Above $Z \approx 40$ the residual rises smoothly to +0.63% at uranium. That is the size and sign expected from the Breit interaction and QED, which reduce K-shell binding by a few hundred eV at uranium and are absent from a Dirac mean field. Below $Z \approx 40$, where those effects are $\lesssim$0.1%, the residual reaches +1.2% (sodium). There it mainly reflects the comparison itself: the Data Booklet references metals to the Fermi level rather than to a free atom's vacuum level, and the transition-state treatment adds its own error. For H and He the K-shell binding is the first ionization energy.

5.3 What the ledger shows

The ledger of eq. (8) has the same shape across the table. The electron that ionizes loses between about one second (lithium's 2s) and 36 seconds (copernicium's 6d) of proper time per day, half to fifteen times the exact hydrogen value of 2.30 s. It dips through the core, where the enclosed charge is large. What makes it loosely held is shelter: the potential of the charge lying outside it cancels about two-thirds to three-quarters of its clock term for the alkali s electrons (66% for Li to 76% for Fr), and up to 97% for palladium's 4d. The exchange–correlation term is not small either; for the ionizing electron its median size is half of $|\varepsilon|$. The binding of the outer electron is therefore a small residue of a large clock term and a large shelter term, with a smaller exchange–correlation term, which is why chemistry is so sensitive.

The ledger also shows what orders calcium's 4s and 3d orbitals. In calcium's field the 3d is

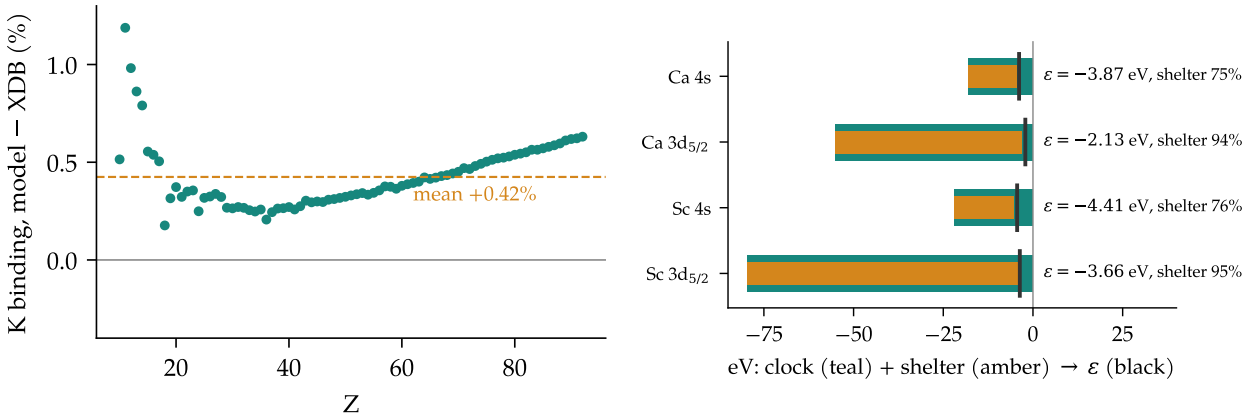


Figure 5: Left: K-shell binding energies against the X-ray Data Booklet. Right: the clock-plus-shelter ledger of the 4s and $3d_{5/2}$ orbitals of calcium and scandium, in the spin-up channel of each neutral atom's self-consistent potential. Calcium's 3d is evaluated unoccupied in calcium's own field.

the *slower* clock of the two. Its clock term, −55.2 eV, is 3.05 times the 4s's −18.1 eV, because it sits deeper, at $\langle r \rangle = 2.3$ against 4.0 bohr. But 94% of that clock is sheltered, against 75% for the 4s, so the 4s orbital lies 1.7 eV lower (Figure 5, right). The ordering of the two orbital energies is set by shelter, not by the clock. Which orbitals are occupied is decided by total energies, not orbital energies alone. In scandium the occupied 3d still lies 0.75 eV above the 4s (spin-up channel), the familiar 4s/3d paradox, which the ledger describes but does not resolve; for its resolution through total energies and the repulsion within the d shell, see [35].

At the heavy end, the 1s clock of oganesson loses 11.3 hours per day. From rutherfordium to copernicium the 7s is pulled inward where clocks are slow. In this model the 6d is the electron that ionizes in every one of them except bohrium, where 7s wins by 0.1 eV. This is a prediction; none of these ions has been measured.

6 Results: the decay clocks

6.1 Spontaneous emission and the cube of the period

For hydrogen's 2p state, eq. (9) with the measured period $T = 0.4055$ fs and $t_d = 0.1315$ as gives 1.598 ns (1.595 ns with the Dirac period), against 1.596 ns from the NIST A-value 6.2649×10^8 s^{-1}. For the alkali resonance lines (Table 2), the dipole comes from the computed atoms and the period from the measured line. Each of the ground and excited states is solved self-consistently. The predicted lifetimes land within +0.6% (Li), +1.1% (Na), −3.9% (K), −8.0% (Rb) and −8.1% (Cs) of the NIST Handbook values [27]. The Handbook's Rb value is older; more recent measurements are closer to 26.2 ns, which would move Rb to about −5% [30]. The heavy-alkali shortfall is core polarization, which a mean field omits. Because the period enters cubed, a 3% error in the line energy becomes a 9% error in the lifetime. That is why the fully computed version, which uses the model's own period, drifts by up to −18%. Figure 6 shows the Z^{-4} law for one-electron ions, from nanoseconds in hydrogen to 7 attoseconds in the heaviest ions. Neutral atoms' valence lifetimes do not follow it; the alkalis all sit at 16–31 ns.

Table 2: The cube law against reference lifetimes. R: radial dipole integral (bohr); T: measured period (fs); $t_d = R/(\sqrt{3}\,c)$ (as); lifetimes in ns; error uses the measured T.

line	R	T	t_d	τ (meas. T)	τ (model T)	τ (NIST)	error %
H $2p_{3/2}$	1.2903	0.4055	0.1315	1.598	1.595	1.596	+0.1
Li $2p_{3/2}$	4.050	2.238	0.413	27.27	28.69	27.10	+0.6
Na $3p_{3/2}$	4.295	1.965	0.438	16.42	14.57	16.23	+1.1
K $4p_{3/2}$	5.130	2.557	0.523	25.35	23.03	26.39	−3.9
Rb $5p_{3/2}$	5.319	2.603	0.542	24.86	22.05	27.03	−8.0
Cs $6p_{3/2}$	5.715	2.843	0.582	28.07	25.64	30.53	−8.1

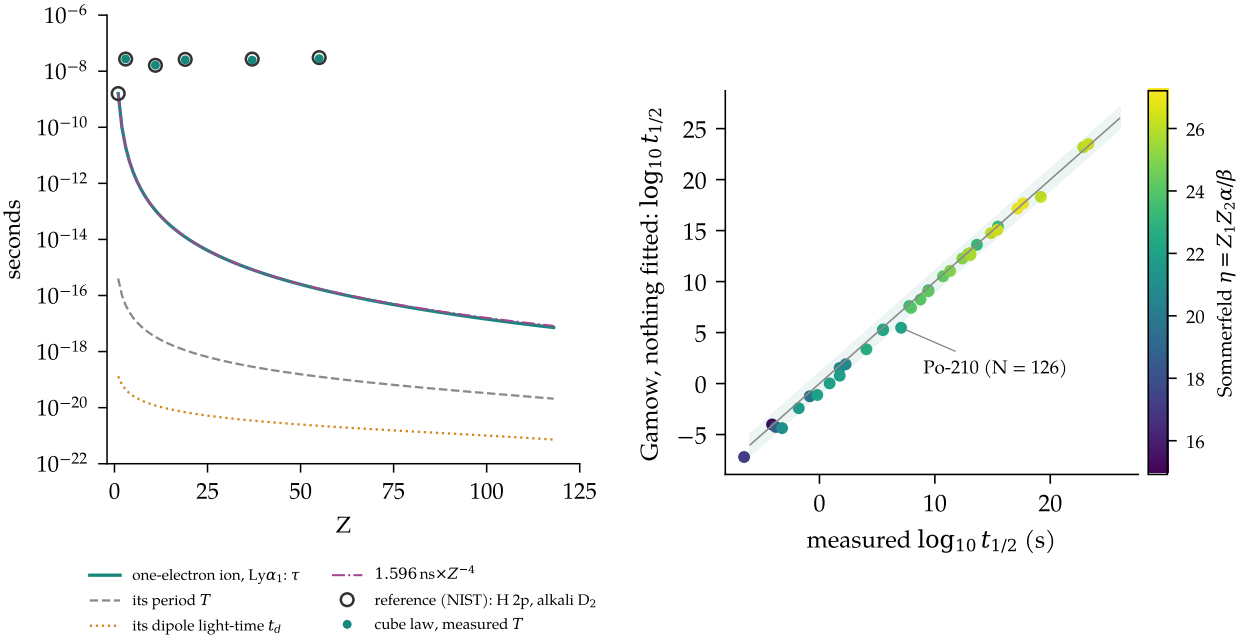


Figure 6: Left: the cube law. One-electron-ion lifetimes, with their periods and dipole times, the Z^{-4} guide, and reference lifetimes (open circles) with cube-law predictions (dots). Right: alpha-decay half-lives from eq. (10) with $r_0 = 1.2$ fm and no fitted constant, against measurement; the band is ± 1 decade.

Where the first edition's t_3 may live. [OPEN] The fine-structure constant α enters eq. (9) exactly once, as the coupling between the period's cube and the dipole's square. That is the role the first edition gave t_3. Its "Scramble Zone", picoseconds to nanoseconds, is also where optical spontaneous emission sits: 1.6 ns for hydrogen, 16–31 ns for the alkalis. The cube is radiation reaction: the force that ends the state is $m\tau_0\, d^3x/dt^3$, the jerk, with $\tau_0 = \frac{2}{3}\alpha t_C$, and every line's quality factor $\omega\tau$ scales as α^{-3} (Section 8.5). Whether the first edition's third time dimension is this clock remains open.

6.2 Alpha decay

Across 35 even-even emitters from Te-106 to Og-294 (Appendix F), eq. (10) with $r_0 = 1.2$ fm and no fitted constant predicts half-lives spanning 29.8 decades. The RMS error is 0.55 decades and the mean offset is -0.39 decades. The absolute level depends on the assumed radius: $r_0 = 1.15$, 1.25 and 1.3 fm give RMS errors of 0.46, 1.11 and 1.70 decades. The scatter about a single overall

offset does not: it stays at 0.37–0.46 decades for r_0 between 1.15 and 1.3 fm. One overall constant, standing for the alpha-formation probability, gives 0.40 decades at $r_0 = 1.2$ fm [FITTED], but the result does not need it. Po-210 is the one outlier, at -1.6 decades raw and -1.2 after the offset. It sits on the $N = 126$ shell closure, where alpha particles are known to form less readily. The nuclear clock $t_a \approx 10^{-21}$ s is of the same order as the electron's Compton period, and each half-life is that tick multiplied by e^{2G}, between 10^{14} and 10^{44} ticks.

6.3 The radioactive edge of the table

Appendix G lists the longest-lived isotope of every element with no stable isotope [MEASURED] [28]. Half-lives run from 2×10^{19} years (Bi-209) down to under a millisecond (Og-294). For the alpha emitters among them, the ratio $Z\alpha/\beta$ that sets an atom's clock also sets the decay: the decay energy Q grows beyond uranium, β grows with it, the Sommerfeld number falls, and $e^{2\pi\eta}$ collapses. Tc, Pm, At, Fr and Ac decay instead by electron capture or β decay, and Lr, Rf and Ds by fission.

7 The atom as a test of the temporal framework

The Chronowave Temporal Framework makes claims about clocks that go beyond this paper's premise, and its interactive companion, the Temporal Engine, already rules on several of them. Two can be settled on the atom with the data used above: the framework's cubic time-dilation law and its static temporal volume. Table 3 maps the framework's claims onto the atom. The rest of this section measures the dilation exponent from one-electron spectra (Section 7.1), separates the atom's slow clocks from gravity's (Section 7.2), locates α among the atom's clocks (Section 7.3) and states what the atom says about the temporal volume (Section 7.4).

Table 3: Framework claims and their atomic counterparts. Module numbers refer to the Temporal Engine.

framework claim	in the atom	verdict here
the temporal atom, $E = mc^2\langle d\tau/dt\rangle$ (Module VIII)	eq. (1), exact for $1/r$ by eqs. (2) and (3)	holds (Sections 3–5)
cubic dilation, $d\tau/dt = (1-\beta^2)^{1/3}$ (Module VI)	a bound electron moving at $\beta = Z\alpha/n$	excluded as a clock law, $\lvert k - \frac{1}{2}\rvert < 1.9 \times 10^{-8}$ (Section 7.1); the exponent recurs in radiation (Section 8.7)
the same law in gravity, $d\tau/dt = (1 - r_s/r)^k$ (Module XI)	atomic and orbiting clocks compared	one law for rates, not for energies (Section 7.2)
α carried by a third time t_3 (first edition)	$\alpha = t_C/(a_0/c)$, a ratio of clocks	[OPEN] (Section 7.3)
a static temporal volume, $n_c = 0$ (Module IX)	$A = V_T^{-1/2}$; the constancy of α	consistent (Section 7.4)

7.1 The clock law, read off the spectrum

The framework's Module VI replaces the square root in the Lorentz factor with a cube root, $\gamma_{\text{cubic}} = (1 - v^2/c^2)^{-1/3}$. Write the family as

$$\frac{d\tau}{dt} = \left(1 - \beta^2\right)^k, \qquad k = \tfrac{1}{2} \text{ (special relativity)}, \qquad k = \tfrac{1}{3} \text{ (cubic)}. \tag{11}$$

Module XI has already measured the gravitational form of the same law, $d\tau/dt = (1 - r_s/r)^k$. Four redshift anchors give $\hat{k} = 0.50000 \pm 0.00004$, which excludes $k = \frac{1}{3}$ at 4762σ. That leaves the kinematic exponent, and the framework's one retreat: $k_{\text{kin}} = \frac{1}{3}$ with $k_{\text{grav}} = \frac{1}{2}$. The Engine closes the retreat indirectly, because clocks on the rotating geoid tick alike only if the two exponents are equal, and caesium fountains bound their difference to 2×10^{-4}. The kinematic exponent has also been measured directly, and far more precisely, with moving clocks: in an Ives–Stilwell experiment, $^7\text{Li}^+$ ions stored at $\beta = 0.338$ confirmed $\gamma\sqrt{1-\beta^2} = 1$ to 2.3×10^{-9} [33]. Under eq. (11) the dilation factor is $(1 - \beta^2)^{-k}$, so that product becomes $(1 - \beta^2)^{1/2-k}$, and

$$|k_{\text{kin}} - \tfrac{1}{2}| < \frac{2.3 \times 10^{-9}}{|\ln(1 - \beta^2)|} = 1.9 \times 10^{-8}. \tag{12}$$

The atom adds an independent measurement inside bound states, at speeds up to $\beta = 0.67$ for the 1s electron of U^{91+} (the GNSS clocks of Module VI move at $\beta \approx 1.3 \times 10^{-5}$), but it is about 5000 times less precise.

Worked example 21: the storage-ring bound on k

$\beta = 0.338$, so $\beta^2 = 0.114244$ and $|\ln(1 - \beta^2)| = 0.121314$. A departure $k - \frac{1}{2}$ shifts $\gamma\sqrt{1-\beta^2}$ by $(k - \frac{1}{2}) \times 0.121314$, and the experiment allows 2.3×10^{-9}: $|k - \frac{1}{2}| < 2.3 \times 10^{-9}/0.121314 = 1.90 \times 10^{-8}$. For the cubic law, $\gamma\sqrt{1-\beta^2} = (1 - \beta^2)^{1/6} = 0.979984$, a 2.0% departure, 8.7×10^6 times what the experiment allows. *Try it:* the same precision at $\beta = 0.064$ would bound k only to 5.6×10^{-7}, because $|\ln(1 - \beta^2)|$ falls as β^2; high speed is what makes the test sharp.

The derivation. Apply the premise $E = E_0\langle d\tau/dt\rangle$ to an electron on the circular Sommerfeld orbit n, whose speed is $\beta = Z\alpha/n$, and let its clock obey eq. (11):

$$E_n = E_0\,(1 - x)^k, \qquad B_n = E_0\left[1 - (1 - x)^k\right], \qquad x = \left(\frac{Z\alpha}{n}\right)^2. \tag{13}$$

At $k = \frac{1}{2}$ and $E_0 = mc^2$ this is exactly the Dirac circular level of eq. (4), so the family contains the true spectrum. For $k \neq \frac{1}{2}$ the rest energy E_0 has to be chosen, and there are two readings.

- *Reading A*, $E_0 = mc^2$. Expanding, $B_n \approx k\,x\,mc^2 = 2k \times B_{\text{Bohr}}$, so the cubic law gives hydrogen an ionization energy of $\frac{2}{3} \times 13.598 = 9.066$ eV. Every measurement of hydrogen excludes it.
- *Reading B*, $E_0 = mc^2/2k$, chosen so that the non-relativistic limit is Bohr's for every k. The price is an electron rest energy of 766.5 keV at $k = \frac{1}{3}$, against the 511 keV of the annihilation line. Grant it anyway. Then only the relativistic part of the spectrum depends on k:

$$\frac{B_n}{B_{\text{Bohr}}} - 1 = \frac{1 - k}{2}\,x + O(x^2), \qquad \frac{\partial B_n}{\partial k} = -\frac{mc^2}{4}\,x^2 + O(x^3). \tag{14}$$

Reading B is the one most favourable to the framework, and every number below uses it.

Worked example 22: how far apart $k=\frac{1}{2}$ and $k=\frac{1}{3}$ put three ions

Reading B, 1s level ($n=1$), reduced masses as in Table 1. *Hydrogen*, $x=\alpha^2=5.3251\times10^{-5}$: $B/B_{\mathrm{Bohr}}-1=1.331319\times10^{-5}$ at $k=\frac{1}{2}$ and 1.775098×10^{-5} at $k=\frac{1}{3}$, close to $x/4$ and $x/3$. The cubic law deepens hydrogen's binding by 4.44 ppm. He^{+}, $x=4\alpha^2$: 5.325703×10^{-5} against 7.101021×10^{-5}, a difference of 17.8 ppm. U^{91+}, $x=(92\alpha)^2=0.450719$: 0.1486763 against 0.2049811. In energy, $B=132\,279.63$ eV at $k=\frac{1}{2}$ and 138 763.60 eV at $k=\frac{1}{3}$. The cubic law moves the level by 6484.0 eV, fourteen times the entire measured Lamb shift (460.2 eV). The lever per unit k from eq. (14) is $-(mc^2/4)x^2=-25\,952$ eV at leading order; the exact derivative is $-37\,652$ eV, because at $x=0.45$ the higher orders matter. *Try it:* for the 1s binding of Ar^{17+}, $x=(18\alpha)^2=0.017253$, and the leading-order lever is $-(510\,999/4)\times0.017253^2=-38.03$ eV per unit k; the exact value is -38.47 eV.

The measurement. For each measured quantity P, either a Lyman-α_1 transition energy or a 1s Lamb shift, the full theory at $k=\frac{1}{2}$ (QED, finite nuclear size and recoil; Yerokhin and Shabaev [23], Table V) is compared with experiment. Whatever residual r remains is charged to a departure of the exponent:

$$\hat{k}-\tfrac{1}{2}=\frac{r}{\partial P/\partial k},\qquad \sigma_k=\frac{\sigma}{|\partial P/\partial k|}. \tag{15}$$

Here σ combines the experimental and theoretical uncertainties in quadrature, and $\partial P/\partial k$ is evaluated exactly in reading B. Table 4 lists 14 measurements from that compilation, Mg^{11+} to U^{91+}. Their weighted mean is

$$\hat{k}=0.499962\pm0.000102,\qquad \chi^2=9.10 \text{ for } 13 \text{ degrees of freedom}.$$

Special relativity's $k=\frac{1}{2}$ lies 0.4σ away; the cubic $k=\frac{1}{3}$ lies 1626σ away (Figure 7).

Table 4: The dilation exponent from each measured one-electron ion. Lyα_1: the $2p_{3/2}$–1s transition energy; Lamb: the 1s Lamb shift. Theory and experiment in eV, as compiled in [23]. r: measured minus predicted line or binding energy, in eV. $\partial P/\partial k$: eV per unit k.

ion	quantity	theory	experiment	r	$\partial P/\partial k$	$\hat{k}-\frac{1}{2}$	σ_k
Mg^{11+}	Lyα_1	1472.63626	1472.671(35)	+0.0347	−7.080	-4.91×10^{-3}	4.94×10^{-3}
S^{15+}	Lyα_1	2622.69961	2622.704(20)	+0.0044	−22.471	-1.95×10^{-4}	8.90×10^{-4}
Cl^{16+}	Lyα_1	2962.35331	2962.49(12)	+0.1367	−28.674	-4.77×10^{-3}	4.18×10^{-3}
Ar^{17+} (2014)	Lyα_1	3322.99315	3322.993(14)	−0.0001	−36.087	$+4.2\times10^{-6}$	3.88×10^{-4}
Ar^{17+} (1985)	Lyα_1	3322.99315	3322.989(17)	−0.0041	−36.087	$+1.15\times10^{-4}$	4.71×10^{-4}
Ti^{21+}	Lamb	2.3319	2.29(14)	+0.0419	−86.346	-4.85×10^{-4}	1.62×10^{-3}
V^{22+}	Lyα_1	5443.6307	5443.95(25)	+0.3193	−96.952	-3.29×10^{-3}	2.58×10^{-3}
Fe^{25+}	Lyα_1	6973.1815	6972.65(75)	−0.5315	−159.212	$+3.34\times10^{-3}$	4.71×10^{-3}
Ni^{27+}	Lyα_1	8101.7494	8101.76(11)	+0.0106	−215.039	-4.9×10^{-5}	5.12×10^{-4}
Ge^{31+}	Lyα_1	10624.5338	10624.24(16)	−0.2938	−370.270	$+7.93\times10^{-4}$	4.32×10^{-4}
Kr^{35+}	Lyα_1	13508.965	13508.95(50)	−0.0150	−599.444	$+2.5\times10^{-5}$	8.34×10^{-4}
Xe^{53+}	Lamb	47.09	54.0(10.0)	−6.91	−3439.6	$+2.01\times10^{-3}$	2.91×10^{-3}
Au^{78+}	Lamb	205.2	202.0(8.0)	+3.2	−18242.8	-1.75×10^{-4}	4.39×10^{-4}
U^{91+}	Lamb	463.8	460.2(4.6)	+3.6	−37651.8	-9.56×10^{-5}	1.23×10^{-4}

Worked example 23: one ion's exponent, and the mean

U^{91+}. Theory gives a 1s Lamb shift of 463.8 ± 0.5 eV and experiment 460.2 ± 4.6 eV, so the measured binding exceeds the predicted one by $r=463.8-460.2=+3.6$ eV. The lever is $\partial B/\partial k=-37\,651.8$ eV, so eq. (15) gives $\hat{k}-\frac{1}{2}=3.6/(-37\,651.8)=-9.56\times10^{-5}$, with $\sigma_k=\sqrt{4.6^2+0.5^2}/37\,651.8=1.23\times$

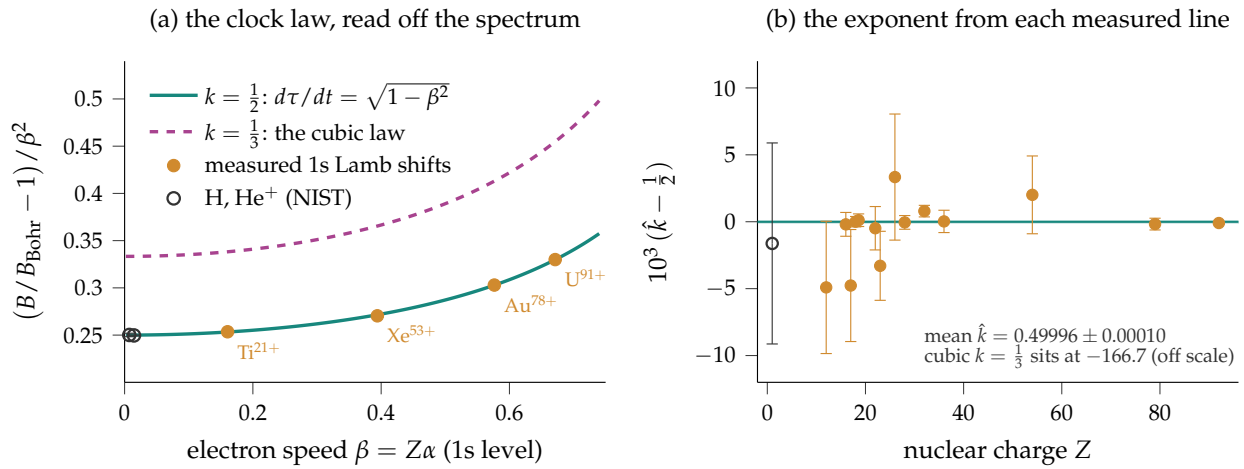


Figure 7: The clock law measured on the atom. (a) The relativistic part of the 1s binding, $(B/B_{\mathrm{Bohr}} - 1)/\beta^2$, against the electron's speed $\beta = Z\alpha$, in reading B. Special relativity ($k = \frac{1}{2}$, teal) starts at $\frac{1}{4}$; the cubic law ($k = \frac{1}{3}$, dashed) starts at $\frac{1}{3}$ and stays above it. Points: measured 1s bindings with the theoretical Lamb shift added back (Ti, Xe, Au and U, filled) and the NIST reference values for H and He$^+$, treated the same way (open). (b) The exponent from each measurement of Table 4 (filled) and from Herzberg's deuterium line (open), with the weighted mean and its $\pm 1\sigma$ band. The cubic law lies at $10^3(\hat{k} - \frac{1}{2}) = -166.7$, far below the panel.

10^{-4}. *Ge*$^{31+}$. Lyα_1 is measured at 10 624.24 ± 0.16 eV against 10 624.5338 eV in theory: $r = -0.2938$ eV, lever -370.27 eV, so $\hat{k} - \frac{1}{2} = +7.93 \times 10^{-4} \pm 4.32 \times 10^{-4}$. *The mean* weights each value by $w_i = 1/\sigma_{k,i}^2$: $\hat{k} - \frac{1}{2} = \sum_i w_i(\hat{k}_i - \frac{1}{2}) / \sum_i w_i$ and $\sigma = (\sum_i w_i)^{-1/2}$. U^{91+} alone has $w = 6.62 \times 10^7$ of the total 9.61×10^7, so it carries 69% of the weight. *Try it:* with only these two ions, $w_{\mathrm{Ge}} = 5.36 \times 10^6$ and $\hat{k} - \frac{1}{2} = \big(6.62 \times 10^7 \times (-9.56 \times 10^{-5}) + 5.36 \times 10^6 \times 7.93 \times 10^{-4}\big)/7.16 \times 10^7 = -2.9 \times 10^{-5}$.

Without QED. The residuals above lean on QED theory. The heaviest ion makes the point without it. Treat U^{91+}'s entire measured Lamb shift, 460.2 eV, as an unknown systematic. Then $\sigma_k = 460.2/37\,652 = 0.0122$, and $k = \frac{1}{3}$ still lies $(1/6)/0.0122 = 13.7\sigma$ away, from one ion and no radiative theory at all.

The light atoms. Hydrogen and He$^+$ give, from the Lamb-corrected residuals of Table 1 with 0.1 ppm allowed for each reference value, $\hat{k} - \frac{1}{2} = -0.0004 \pm 0.0038$ and $+0.0008 \pm 0.0009$. Their levers are small, -26.6 and -106.5 ppm of the binding per unit k. Herzberg's 1956 deuterium line alone gives $\hat{k} - \frac{1}{2} = -0.0016 \pm 0.0075$, so the cubic law was already excluded at 22σ by a measurement made seventy years ago.

7.2 Two kinds of slow clock

The framework writes kinematic and gravitational dilation with one law. As *rates* they do obey one law, with $k = \frac{1}{2}$ in both channels. As sources of *energy* they differ, and the atom shows how. For an electron in any one-electron atom, eq. (1) makes the energy proportional to the clock rate, so binding equals the clock deficit: $1 - E/mc^2 = 1 - f$, with $f = \langle d\tau/dt \rangle$. For a mass on a circular orbit around a Schwarzschild mass M, with $u = GM/rc^2$, the clock rate and the energy per unit

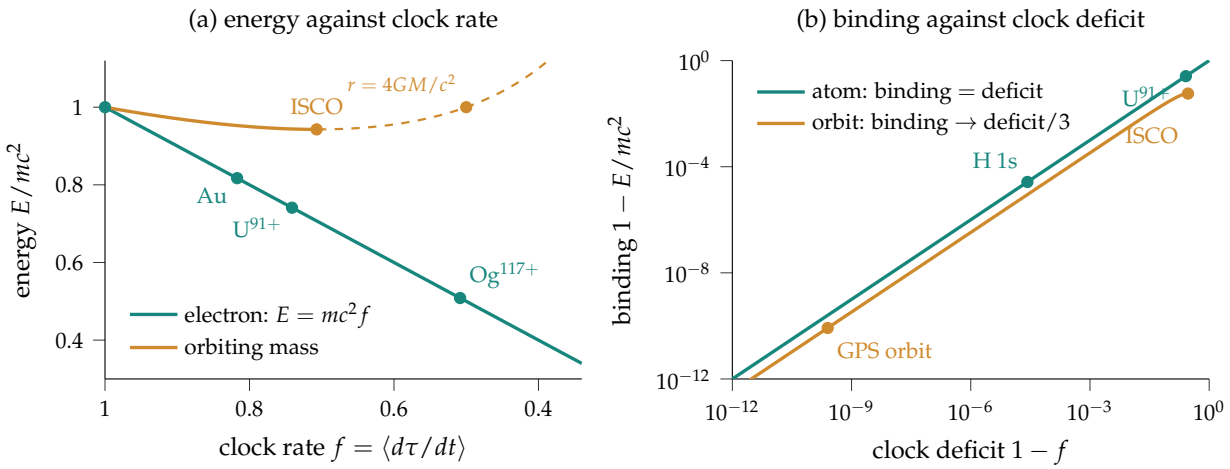


Figure 8: Two kinds of slow clock. (a) Energy against clock rate. An electron in a one-electron atom lies on $E = mc^2 f$ (teal; the 1s levels of H, Au, U^{91+} and Og^{117+} are marked). A mass on a circular Schwarzschild orbit follows eq. (16) (amber). Its energy falls to the ISCO and rises beyond it (dashed, unstable), returning to $E = mc^2$ at $r = 4GM/c^2$ with its clock at half speed. (b) Binding against clock deficit. The atom lies on the diagonal; the orbit lies a factor of three below it in the weak field (GPS orbit) and further below near the ISCO.

rest energy are

$$f = \frac{d\tau}{dt} = \sqrt{1-3u}, \qquad \frac{E}{mc^2} = \frac{1-2u}{\sqrt{1-3u}} = \frac{1+2f^2}{3f}. \tag{16}$$

In the weak field $f \approx 1 - \frac{3}{2}u$ while $E/mc^2 \approx 1 - \frac{1}{2}u$, so the binding is only a third of the deficit. The orbiting clock is slow for two reasons: its motion ($\frac{1}{2}\beta^2 = \frac{1}{2}u$) and its depth in the potential (u). Only the motion is paid for as binding; the depth slows every clock at that height, bound or not. The electron has no depth term, because the Coulomb potential slows no clock, so its whole deficit is motion. The premise $E = mc^2\langle d\tau/dt\rangle$ is therefore a statement about the Coulomb force in flat spacetime, not about slow clocks in general (Figure 8). In the strong field the two part further. At the innermost stable circular orbit (ISCO) the clock runs at 71% while the binding is 5.7%, and at $r = 4GM/c^2$ a clock running at half speed carries no binding at all.

Worked example 24: a GPS satellite and a hydrogen atom, clock for clock

GPS orbit, $r = 26\,560$ km, $GM_\oplus = 3.986004418 \times 10^{14}\ \mathrm{m^3/s^2}$: $u = GM/rc^2 = 3.986004 \times 10^{14}/(2.6560 \times 10^7 \times 8.987552 \times 10^{16}) = 1.669815 \times 10^{-10}$. Relative to a distant static clock, eq. (16) gives a deficit $1 - \sqrt{1-3u} = 2.5047 \times 10^{-10}$ (21.6 μs per day) and a binding $1 - (1-2u)/\sqrt{1-3u} = 8.3491 \times 10^{-11}$. Their ratio is 0.3333. *Hydrogen 1s:* the deficit and the binding are both 2.662603×10^{-5} of mc^2. Their ratio is exactly 1. *The proton's own gravity* at the Bohr radius slows the electron's clock by $Gm_p/(a_0c^2) = 2.35 \times 10^{-44}$, 10^{39} times less than the electron's motion does. *A static clock in gravity with the atom's rate:* $\sqrt{1-r_s/r} = f$ gives $r = r_s/(1-f^2)$. For hydrogen's 1s rate, $1 - f^2 = \alpha^2$ and $r = r_s/\alpha^2 = 18\,779\,r_s$; for U^{91+}'s, $r = 2.2187\,r_s$, inside the ISCO at $3\,r_s$. *Try it:* at the ISCO, $u = \frac{1}{6}$: $f = \sqrt{1/2} = 0.70711$ and $E/mc^2 = (2/3)/\sqrt{1/2} = 0.94281$, so the clock loses 29.3% while the binding is 5.72%.

7.3 Where α lives in time

The first edition gave the fine-structure constant its own time dimension, t_3. In the atom α is not a clock but a ratio of clocks. The Compton time $t_C = \hbar/mc^2$ is the electron's rest clock. The light-crossing time of the Bohr radius, a_0/c, and the atomic unit of time, $t_{\text{au}} = \hbar/E_h$, are the atom's:

$$\frac{t_C}{a_0/c} = \alpha, \qquad \frac{t_C}{t_{\text{au}}} = \alpha^2, \qquad \langle d\tau/dt \rangle_{1s}^{\text{H}} = \cos\theta \ \text{ with } \ \sin\theta = \alpha. \tag{17}$$

The last form reads hydrogen's clock rate as a tilt. Since $(c\,d\tau)^2 + (dx)^2 = (c\,dt)^2$, proper time and motion through space are the two legs of a right triangle whose hypotenuse is coordinate time, and hydrogen's ground-state clock is tilted by $\theta = \arcsin\alpha$ from the laboratory's [FORCED-BY-DEFINITION]. Whether this is the role the first edition meant for t_3 remains [OPEN]. The sensitivity of each line to α, which only the proper-time part of the line can carry, is worked in the Discussion.

Worked example 25: α from three clocks

$t_C = \hbar/mc^2 = 6.582119569 \times 10^{-16}/510\,998.95069 = 1.288089 \times 10^{-21}$ s. $a_0/c = 5.29177210544 \times 10^{-11}/299\,792\,458 = 1.765145 \times 10^{-19}$ s. $t_{\text{au}} = \hbar/E_h = 6.582119569 \times 10^{-16}/27.211386246 = 2.418884 \times 10^{-17}$ s. The ratios are $1.288089 \times 10^{-21}/1.765145 \times 10^{-19} = 7.29735 \times 10^{-3} = \alpha$ (so $1/\alpha = 137.036$) and $1.288089 \times 10^{-21}/2.418884 \times 10^{-17} = 5.32514 \times 10^{-5} = \alpha^2$. The tilt is $\theta = \arcsin(0.0072973526) = 0.0072974$ rad $= 0.41811°$, and $\cos\theta = 0.999973374$ is hydrogen's 1s rate. *Try it:* for a hydrogen-like ion the tilt is $\arcsin(Z\alpha)$: $42.17°$ for U^{91+}, and $90°$, a stopped clock, at $Z\alpha = 1$.

7.4 A static temporal volume

The framework's Module IX lets n_c of the three time dimensions grow with cosmic age, so that $V_T \propto t^{n_c}$, and adopts $n_c = 0$, a static temporal volume. The observed dark-energy equation of state, $w = -0.995 \pm 0.023$ from DESI DR2 with the CMB and Pantheon+ supernovae [32], gives $n_c = 2(w+1) = 0.01 \pm 0.05$; the Union3 and DES-SN5YR samples give 0.01 ± 0.05 and 0.06 ± 0.04. The atom supplies two further routes to the same answer.

1. *Normalization.* The first edition's cure for its divergent normalization (A1) is $A = V_T^{-1/2}$. A cell that grew with cosmic age would make the normalization of every atomic state drift; A is a constant only if V_T is.
2. *The constancy of α.* If a time dimension tracked cosmic age and α depended on it, with $S = d\ln\alpha/d\ln t_i$, then α would drift at $\dot\alpha/\alpha = S/t_0$. Optical clocks bound the drift, and only the proper-time part of a line can register it: in the non-relativistic atom every line scales as α^2, which is indistinguishable from a change of units.

Worked example 26: the bound on a time-dependent α

Clock comparisons give $\dot\alpha/\alpha = (1.0 \pm 1.1) \times 10^{-18}$ yr^{-1} [31]; at 2σ, $|\dot\alpha/\alpha| < 1.0 \times 10^{-18} + 2 \times 1.1 \times 10^{-18} = 3.2 \times 10^{-18}$ yr^{-1}. A time coordinate that tracks cosmic age changes at $1/t_0 = 1/(13.8 \times 10^9\ \text{yr}) = 7.25 \times 10^{-11}$ yr^{-1}. So $|S| < 3.2 \times 10^{-18}/7.25 \times 10^{-11} = 4.4 \times 10^{-8}$: α is independent of any cosmically growing time to better than one part in twenty million. *Try it:* for $n_c = 3$, the alternative Module IX excludes at 51σ, all three times grow and $V_T \propto t^3$. With $A = V_T^{-1/2}$ the normalization would then fall by $\frac{3}{2} \times 7.25 \times 10^{-11} = 1.1 \times 10^{-10}$ per year.

Can the count change? A count can change. The smallest change, one time dimension starting to carry the cosmic age at some epoch, makes the dark-energy density constant before that epoch

and proportional to $1/t$ after it, so w jumps from -1 to $-1 + 1/(3Ht) \approx -0.64$. Section 8.9 tests it against DESI DR2 with thresholds fixed before fitting, and it fails them. With the CMB it improves χ^2 by only 0.5, against 7.9 for DESI's two-parameter fit, and a switch-on more than 4.5 Gyr ago is excluded at 3σ. The supernova samples favour a very late switch-on, between $z = 0.04$ and 0.15, and DES's recalibrated sample halves that preference. No count of time dimensions can give $w < -1$, which DESI's fits reach near $z \approx 0.5$; a confirmed crossing would exclude the whole family.

The partition. The atom confirms the framework's structural claims and excludes its dynamical one. The temporal atom holds with no free parameter, and a static temporal volume is consistent with it. The cubic exponent was already excluded for gravitational clocks by the Engine's own redshift anchors; for moving clocks it is excluded by storage-ring time-dilation measurements, $|k - \frac{1}{2}| < 1.9 \times 10^{-8}$, and independently inside atoms at speeds up to $0.67c$. The Engine left open whether the cubic exponent is a claim about clocks at all, or a bookkeeping convention inside the temporal relabelling. As a claim about clocks it is excluded in both channels. The exponent does recur in radiation, though not for the framework's reasons: once the photons an electron emits carry a large share of its energy, the power it radiates in a fixed strong field grows as $(1 - \beta^2)^{-1/3}$, an exponent set by the Airy structure of the emission. Cubic time also appears wherever an orbit weighs a mass, $GM/c^3 = (a/c)^3(2\pi/P)^2$; wherever radiation reaction (the jerk) acts; and wherever a path bends, making the arrival time cubic in the emission time. Section 8 tests each against measurement.

8 Where cubic time lives

Section 7 found that the framework's cube root fails as a clock rate. This section asks where a cube of time does contribute, and tests each answer against measurement. Kepler's law makes mass a cubic time (Section 8.3). Binary-pulsar timing measures a second gamma built on the cube root of a 2π-normalized period (Section 8.4). The radiation-reaction force is the jerk, and its band is the framework's Scramble Zone (Section 8.5). A bent path makes the arrival time of light cubic (Section 8.6). The framework's exponent recurs, by coincidence, in quantum-limit radiation (Section 8.7), and the cosmic jerk tests the static temporal volume (Section 8.8). A pre-registered test of the smallest change the framework allows, one time dimension switching on, follows (Section 8.9). Section 8.10 lists what to change in the framework.

8.1 The framework's cube-root claims

The framework's cubic-time content is spread over five documents: the 416-page compendium (TF), the GNSS proposal (TF pp. 2–52), the 3-D gravitational-potentials paper (GP, also TF pp. 115–223) and the dark-energy paper (DE, also TF pp. 281–416). Five claims carry the weight:

C1. $\gamma_{\text{cubic}} = (1 - v^2/c^2)^{-1/3}$ as the time-dilation factor of a moving clock, said to be "validated by 1,000 simulations" (TF 2–10). No simulation output, metric or clock data is given.

C2. A third time dimension t_3 that carries α and defines a picosecond-to-nanosecond "Scramble Zone" (TF 238; GP 1–2, 60–72).

C3. A cubic phase $\varphi = \omega t + \alpha t^3$, called "the cubic time dilation model" (GP 21–29).

C4. Field-equation terms $(\epsilon_3/\sqrt{2})\left[\alpha_3(t^3/\tau)^2 - \beta_3(dt^3/d\tau)^2\right]$ (DE 37).

C5. A dark-energy density $\rho_\Lambda = \alpha_3/t^3$ (DE 64–99), and, in the Temporal Engine, the static temporal volume $n_c = 0$.

The framework contains no radiation, jerk, or third-derivative physics. That gap turns out to be where cubic time lives.

The intuition being tested. The question that led here was put in words, not equations. Table 5 records how each phrase was turned into something testable, and where it landed.

Table 5: From the intuition to the physics it pointed at.

the intuition	where it lands (section)
"If a vector becomes geodesic, the cubic vector will also bend"	the jerk of a curved worldline turns with the velocity, $\dddot{\mathbf{x}} = -\omega^2\mathbf{v}$ on a circle; on a geodesic in curved spacetime the jerk term is replaced by a curvature term set by local mass-energy (§8.5)
"an amplitude rarely if ever measured temporally"	$\tau_0 = 6.27 \times 10^{-24}$ s has never been measured as a duration; the Einstein delay γ_E is measured, in seconds, in only a handful of binaries (§8.5, §8.4)
"fluctuates as jerk, normally defined as a quantity or quality over time"	radiated power (energy per time), linewidth, and the quality factor $Q = \omega\tau$ (§8.5)
"cubic time must account for something like mass"	$GM/c^3 = (a/c)^3(2\pi/P)^2$: mass is a cubic time (§8.3)
"or the beat time's curvature"	a bent path makes the arrival time cubic (§8.6)
"a second gamma"	γ^3 compresses the pulse from a bent path; pulsar timing's γ_E (§8.6, §8.4)
"a second normalized gamma of T^3 and t or τ, normalized to 2π"	$\beta^3 = T_M/(P/2\pi)$; $\gamma_E \propto (P_b/2\pi)^{1/3}$; $Q/2\pi = \tau/T \propto \alpha^{-3}$ (§8.3, §8.4, §8.5)
"just a different direction"	the reaction force is tangential, perpendicular to the binding force, tilted by $\frac{2}{3}\alpha^3$ rad (§8.5)

8.2 The cube root on the GNSS proposal's own satellites

Give motion and gravity separate exponents, as the Temporal Engine does:

$$\frac{d\tau}{dt} = (1-\beta^2)^{k_{\text{kin}}}\left(1 - \frac{r_s}{r}\right)^{k_{\text{grav}}} \approx 1 - k_{\text{kin}}\frac{v^2}{c^2} - 2k_{\text{grav}}\frac{GM}{rc^2}. \tag{18}$$

GR is $(\frac{1}{2}, \frac{1}{2})$; the framework's GNSS law is $(\frac{1}{3}, \frac{1}{3})$; the Engine's "retreat" keeps the cube root for motion only, $(\frac{1}{3}, \frac{1}{2})$. The GNSS proposal's own satellite table includes Galileo 5 and 6 (GSAT0201/0202, eccentricity 0.16), whose clocks were later used for the most precise redshift test made with satellites [37].

Worked example 27: GPS, NTS-2 and Galileo against eq. (18)

GPS. With $a = 2.656175 \times 10^7$ m, $GM/(ac^2) = 1.669705 \times 10^{-10}$; the geoid constant is $\Phi_0/c^2 = 6.969290134 \times 10^{-10}$ [38]. A satellite clock runs fast against a ground clock by $2k_{\text{grav}}\Phi_0/c^2 - (k_{\text{kin}} + 2k_{\text{grav}})GM/(ac^2)$. GR gives 4.464733×10^{-10}, the offset built into every GPS clock

(10.22999999543 MHz). The cubic law gives 2.976489×10^{-10}: used in GPS it would drift 12.86 μs, or 3.85 km of range, per day. The retreat gives 4.743017×10^{-10}, +2.40 μs per day. *NTS-2 (1977).* Measured $+442.5 \times 10^{-12}$ against GR's $+446.5 \times 10^{-12}$, a test at about 1% [38]. The cubic law predicts 297.7×10^{-12} (32σ away); the retreat 474.3×10^{-12} (7σ). *Galileo 5 and 6.* The orbit's eccentricity makes the clock rate swing once per orbit. On a Kepler orbit the periodic parts of $v^2/2c^2$ and GM/rc^2 are equal, one unit each in GR, so eq. (18) gives $2k_{\text{kin}} + 2k_{\text{grav}}$ units. Delva et al. hold the velocity term at special relativity and fit $(1+\alpha)$ to the gravitational one; they find $\alpha = (+0.19 \pm 2.48) \times 10^{-5}$. A clock obeying eq. (18) reads as $\alpha = 2k_{\text{kin}} + 2k_{\text{grav}} - 2$: $-\frac{2}{3}$ for the cubic law ($26\,882\sigma$) and $-\frac{1}{3}$ for the retreat ($13\,441\sigma$). With $e = 0.162$ the swing is about 0.37 μs; the cubic law would remove about 0.12 μs of it from a signal measured to about 5 ps. *Try it:* any law with $k_{\text{kin}} + k_{\text{grav}} \neq 1$ fails here by $|k_{\text{kin}} + k_{\text{grav}} - 1|/1.24 \times 10^{-5}$ standard deviations.

The kinematic exponent is pinned hardest by a storage-ring Ives–Stilwell experiment: $^7\text{Li}^+$ ions at $\beta = 0.338$ confirm $\gamma\sqrt{1-\beta^2} = 1$ to 2.3×10^{-9} [33], so $|k_{\text{kin}} - \frac{1}{2}| < 2.3 \times 10^{-9}/|\ln(1-\beta^2)| = 1.9 \times 10^{-8}$. With the atomic measurement of Section 7.1 ($k_{\text{kin}} = 0.49996 \pm 0.00010$) and the Engine's redshift anchors ($k_{\text{grav}} = 0.50000 \pm 0.00004$), the clock exponent is $\frac{1}{2}$ in both channels. The GNSS proposal's arithmetic has independent slips as well: its worked example overstates both corrections threefold ($\frac{1}{2} \cdot 10^{-10} = 5 \times 10^{-11}$, not 1.5×10^{-10}), its higher series coefficients for $(1-x)^{-1/3}$ should be 14/81 and 35/243, and its gravity term 2.793×10^{-10} should be 1.59×10^{-10}.

8.3 Mass is a cubic time

For a circular orbit, $\omega^2 a^3$ is the coupling of the force. Divide by c^3:

$$T_M \equiv \frac{GM}{c^3} = \left(\frac{a}{c}\right)^3 \left(\frac{2\pi}{P}\right)^2, \qquad \beta = \frac{v}{c} = \left(\frac{T_M}{P/2\pi}\right)^{1/3}. \tag{19}$$

A mass is the cube of a light-crossing time divided by the square of a 2π-normalized period, and an orbital speed is the cube root of a ratio of two times. The same law holds for the electron with $T_M \to r_e/c = \alpha t_C$. In Schwarzschild coordinates $\omega^2 = GM/r^3$ is exact, so the clock of a circular orbit is exactly

$$\frac{d\tau}{dt} = \sqrt{1 - 3\,GM/rc^2} = \sqrt{1 - 3\,(T_M\,\omega)^{2/3}}. \tag{20}$$

The exponent $\frac{2}{3}$ is real, but it sits on mass-time $\times$ frequency, inside GR's square root, not on $(1-\beta^2)$. In gravitational-wave physics this combination is the standard post-Newtonian expansion parameter, $x = (GM\omega/c^3)^{2/3}$ [43].

Worked example 28: weighing the Sun, the Earth and the electron in seconds

Sun. $a = 1$ AU $= 1.495978707 \times 10^{11}$ m, so $a/c = 499.005$ s; the sidereal year is $P = 3.155815 \times 10^7$ s. Then $(a/c)^3(2\pi/P)^2 = 4.925505 \times 10^{-6}$ s. Kepler weighs Sun, Earth and Moon together: removing their 1/328 900.56 leaves 4.925490×10^{-6} s, against $GM_\odot/c^3 = 4.925491$ μs (residual 1.1×10^{-7}). *Earth.* $GM_\oplus/c^3 = 14.7937$ ps. A GPS orbit, half a sidereal day, has $a = 26\,561.76$ km and $\beta = (T_\oplus\,\omega)^{1/3} = 1.29217 \times 10^{-5}$ (3873.83 m/s). *Electron.* For every Bohr orbit $(a_n/c)^3\omega_n^2 = 9.399637 \times 10^{-24}$ s $= r_e/c$, and $\beta_n = (\omega_n r_e/c)^{1/3} = \alpha/n$. At $n = 1$, $\alpha^3 = (r_e/c)/t_{\text{au}} = 3.885939 \times 10^{-7}$: the fine-structure constant is the cube root of a ratio of two of the atom's clocks. *Try it:* eq. (20) gives the innermost stable orbit where $3(T_M\omega)^{2/3} = \frac{1}{2}$, an orbital frequency $1/(2\pi\,6^{3/2}T_M) = 2198.6$ Hz $\times\, M_\odot/M$, and gravitational waves at twice that.

8.4 A second gamma: pulsar timing

With $\beta_M = (T_\odot M/(P_b/2\pi))^{1/3}$ from eq. (19), every relativistic timing parameter of a binary pulsar is a power of one cube root [39]:

$$\frac{\Delta\omega}{\text{orbit}} = 2\pi\,\frac{3\beta_M^2}{1-e^2}, \qquad \gamma_E = e\,\frac{P_b}{2\pi}\,\beta_M^2\,\frac{m_c(m_p+2m_c)}{M^2}, \qquad \dot{P}_b = -\frac{192\pi}{5}\,f(e)\,\frac{m_p m_c}{M^2}\,\beta_M^5. \tag{21}$$

The Einstein delay γ_E is the second gamma: the amplitude, in seconds, of the pulsar clock's slow-down as it moves and falls around an eccentric orbit, measured temporally and in only a handful of systems. Here $f(e) = (1+\frac{73}{24}e^2+\frac{37}{96}e^4)(1-e^2)^{-7/2}$, and the third formula is gravitational radiation, the jerk of the mass distribution (§8.5).

Worked example 29: PSR B1913+16 and the double pulsar

B1913+16 [39]: $P_b = 0.322997448918$ d, $e = 0.6171340$, $\dot\omega = 4.226585^\circ/\text{yr}$, $\gamma_E = 4.307$ ms. The periastron advance fixes $M = 2.8284\ M_\odot$; then γ_E fixes $m_c(M+m_c)$, so $m_p = 1.4379$ and $m_c = 1.3905\ M_\odot$. Now $\beta_M = 1.463807\times10^{-3}$ (439 km/s), and eq. (21) predicts, with nothing free, $\dot{P}_b = -2.40263\times10^{-12}$. The measured intrinsic value is $-(2.398\pm0.004)\times10^{-12}$: ratio 0.9981 ± 0.0017. *J0737−3039A/B* [40]: with $m_A = 1.338185$, $m_B = 1.248868\ M_\odot$, $P_b = 0.1022515592973$ d and $e = 0.087777023$, eq. (21) gives $\gamma_E = 0.383997$ ms against 0.384045 ± 0.000094 measured, and $\dot{P}_b = -1.247810\times10^{-12}$ against -1.247920×10^{-12} observed (ratio 1.00009 before the 10^{-4} kinematic corrections; the paper confirms the radiation to 1.3×10^{-4}). *Try it:* $\beta_M^3 = T_M/(P_b/2\pi) = 13.93\ \mu\text{s}/4441.5\ \text{s} = 3.136\times10^{-9}$ for B1913+16; doubling the period lowers β_M by $2^{1/3}$ and the decay rate by $2^{5/3} = 3.17$.

8.5 The jerk carries cubic time

A charge that accelerates loses energy to radiation, and the force that pays for it is proportional to the third time derivative of position (Abraham–Lorentz [44]):

$$\mathbf{F}_{\text{rr}} = m\,\tau_0\,\dddot{\mathbf{x}}, \qquad \tau_0 = \frac{2}{3}\frac{r_e}{c} = \frac{2}{3}\,\alpha\,t_C = 6.2664\times10^{-24}\ \text{s}. \tag{22}$$

This is the rung one power of α below the Compton clock on the ladder of Sections 7.3 and 9: τ_0, $r_e/c = \alpha t_C$, t_C, $a_0/c = t_C/\alpha$, $t_{\text{au}} = t_C/\alpha^2$. The first edition gave α its own time, t_3; the clock through which α enters dynamics is τ_0. In relativistic form the jerk splits along the worldline's own frame (Frenet–Serret, u the four-velocity, κ_1 the curvature, κ_2 the torsion):

$$\dot{a} = \kappa_1^2\,u + \dot\kappa_1\,e_1 + \kappa_1\kappa_2\,e_2. \tag{23}$$

The first term points along time; only the other two radiate. Uniform acceleration ($\dot\kappa_1 = \kappa_2 = 0$) feels no reaction at all. Circular motion feels only the torsion term, along e_2: a drag in a different direction from the binding. On a geodesic in curved spacetime $\kappa_1 = 0$, but the self-force keeps a curvature term, $\frac{1}{3}q^2(R^\mu{}_\nu u^\nu + u^\mu R_{\nu\lambda}u^\nu u^\lambda)$ [41], and through Einstein's equations $R_{\mu\nu}$ is local mass-energy: the cubic term bends with the geometry. For a mass distribution the same order of derivative radiates gravitationally, $L = (G/5c^5)\langle\dddot{Q}_{ij}\dddot{Q}_{ij}\rangle$, which is the $\dot{P}_b$ of §8.4.

Worked example 30: the jerk on hydrogen's Bohr orbit

$\omega = 1/t_{\text{au}} = 4.134137\times10^{16}$ rad/s, $v = \alpha c$, so $|\dddot{\mathbf{x}}| = \omega^2 v = 3.739\times10^{39}$ m/s^3, pointing along $-\mathbf{v}$. The reaction force is 2.134×10^{-14} N; the Coulomb force is 8.239×10^{-8} N, radial. Their ratio is $\tau_0\omega = \frac{2}{3}\alpha^3 = 2.5906\times10^{-7}$: the total force tilts off radial by that angle. The power, $m\tau_0 a^2 = 4.669\times10^{-8}$ W,

is Larmor's. The orbit's quality factor, $2\pi|E|$ over the energy lost per turn, is $1/(2\omega\tau_0) = \frac{3}{4}\alpha^{-3} = 1.93 \times 10^6$. Spiralling in from a_0 takes $(a_0/c)^3/4(r_e/c)^2 = t_C/4\alpha^5 = 15.56$ ps: a cube of one time over the square of another, the same shape as the lifetime law of eq. (9). *Try it:* on a circle $\mathbf{v} \cdot \mathbf{a} = 0$, so all the reaction work is radiated; on a straight-line oscillation $\mathbf{v} \cdot \dot{\mathbf{a}}$ contains $d(\mathbf{v} \cdot \mathbf{a})/dt$, an energy that sloshes back and forth each cycle and averages to zero (the Schott term).

Worked example 31: quantum lifetimes are jerk damping

The classical damping rate of an oscillating electron is $\tau_0\omega^2$. For Lyman-α ($\omega = 1.549458 \times 10^{16}$ rad/s) it is 1.504455×10^9 s^{-1}; the quantum rate is $A = 6.264903 \times 10^8$ s^{-1}, so $A/\tau_0\omega^2 = 0.416423$, the oscillator strength. In general $A = 3(g_l/g_u)\, f\, \tau_0\omega^2$ for a line from level u to level l, and the weight factor is $3 \cdot \frac{2}{6} = 1$ here: every line's rate is its strength times the jerk damping, up to statistical weights. Since $\omega \propto \alpha^2$ and $A \propto \alpha^5$, the quality factor $Q = \omega/A = 2\pi\tau/T$ scales as α^{-3}:

line	2p	3p	4p	5p	6p	7p	8p
lifetime part (ns)	1.596	5.979	14.666	29.091	50.688	80.894	121.141
$Q\,\alpha^3$	9.611	42.667	110.38	224.20	395.63	636.17	957.34

Try it: double α in the formulas and Q falls by exactly 8 for every line.

The framework's Scramble Zone, "picosecond to nanosecond", 1 ps to 100 ns (claim C2), is this band (Figure 9): the classical collapse at 15.6 ps and the Lyman lifetimes from 1.6 to 121 ns. Its location is right; its mechanism is radiation reaction, not a quantum-to-classical transition of gravity.

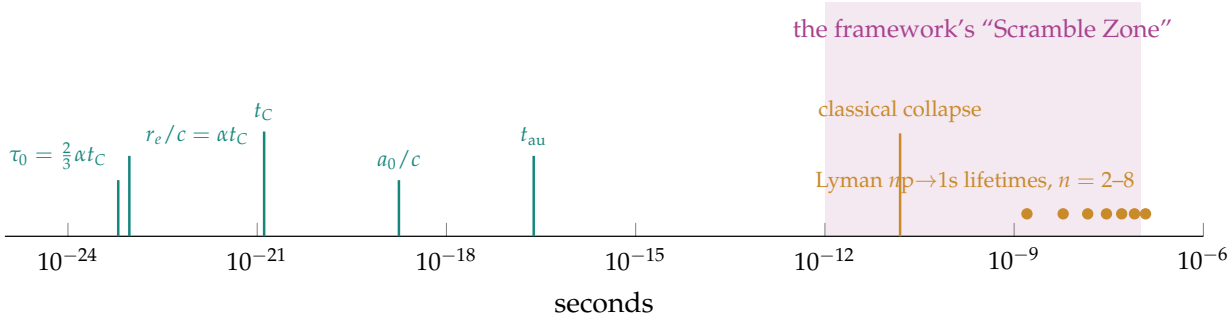


Figure 9: The atom's clocks with the cubic-time rungs: $\tau_0 = \frac{2}{3}\alpha t_C$ and $r_e/c = \alpha t_C$ below the Compton clock, the classical collapse time, and the radiative lifetimes of the Lyman lines, against the framework's Scramble Zone.

8.6 A bent path makes time cubic

Light leaves a moving charge at emission time t and arrives at $t - \mathbf{n} \cdot \mathbf{r}(t)/c$. When the path bends with radius ρ, passing the observer's direction at $t = 0$,

$$t_{\text{obs}} = \frac{t}{2\gamma^2} + \frac{c^2 t^3}{6\rho^2} + \dots \tag{24}$$

The arrival time is cubic in the emission time: the framework's $\varphi = \omega t + \alpha t^3$ (claim C3), with the coefficient fixed by the curvature. The two terms balance at $t \sim \sqrt{3}/\gamma \cdot \rho/c$, so the observed pulse lasts about $\rho/c\gamma^3$, compressed by a second gamma beyond the familiar γ^2 of the Doppler effect. Integrals over a cubic phase are Airy functions, and their spectra are modified Bessel functions of

order $\frac{1}{3}$ and $\frac{2}{3}$, as Schwinger showed for synchrotron light [45]. To test this without assuming it, the radiation field of a charge on a circle was computed directly from the retarded (Liénard–Wiechert) fields and Fourier transformed (Figure 10).

Worked example 32: what the direct integration found

The pulse width times γ^3 is $0.3000\,\rho/c$ for $\gamma = 16$, 32 and 64: a fitted exponent of -3.0000. The spectral peak scales as $\gamma^{3.0023}$, at $\omega_c = \frac{3}{2}\gamma^3 c/\rho$ times a constant. The computed spectrum equals the Bessel form within 0.986–1.002 at four observation angles over $\omega_c/300$ to $4\omega_c$. Below the peak it rises as $\omega^{0.672}$ (theory $\frac{2}{3}$). Integrated over angles it gives $x\int_x^\infty K_{5/3}$ with $x = \omega/\omega_c$, whose low-frequency law is $2^{2/3}\Gamma(\frac{2}{3})\,x^{1/3} = 2.1495\,x^{1/3}$. *Try it:* at $\gamma = 64$, the cubic term of eq. (24) at $t = 1/\gamma$ is exactly 25% of t_{obs}.

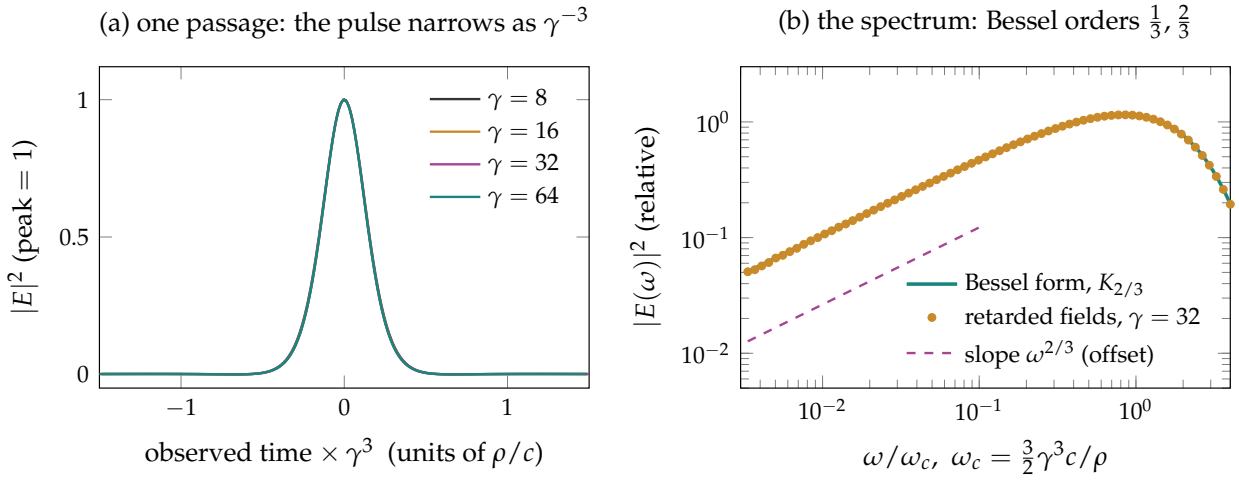


Figure 10: Retarded fields of a charge on a circle. (a) Pulses for $\gamma = 8$–64 collapse onto one curve when time is scaled by γ^3. (b) The computed spectrum (dots) against the Bessel form (line); the dashed line shows the $\omega^{2/3}$ slope, offset for visibility.

8.7 The framework's exponent, by coincidence

When a single emitted photon can carry a large share of the electron's energy (quantum parameter $\chi = \gamma E_\perp/E_S \gtrsim 1$, with E_S the Schwinger field), recoil limits the emission. The radiated power, computed from the full quantum spectrum, grows as χ^n with $n = 2$ classically and $n \to \frac{2}{3}$ for $\chi \gg 1$, where $P \to (32\Gamma(\frac{2}{3})/243)(3\chi)^{2/3}\alpha m^2c^4/\hbar$. At a fixed field $\chi \propto \gamma$, so

$$P \propto \gamma^{2/3} = (1-\beta^2)^{-1/3} \qquad (\chi \gg 1), \tag{25}$$

the same exponent as γ_{cubic}. The match is one of exponents, not of mechanism: the $\frac{2}{3}$ comes from the Airy structure of emission from a bent path (§8.6), not from any argument the framework makes. It marks the one place where a $(1-\beta^2)^{-1/3}$ law is physical, and it does not support the framework's clock law. The same $\frac{2}{3}$ power sets the effective coupling of strong-field QED, $\alpha\chi^{2/3}$, which reaches 1 at $\chi = \alpha^{-3/2} = 1604$ [42]; that the cube root there multiplies α echoes the first edition's guess for t_3, again by coincidence.

Worked example 33: how fast the exponent slides

The local exponent $n = d\ln P/d\ln\chi$ is 1.994 at $\chi = 10^{-3}$, 1.204 at 1, 0.856 at 10, 0.717 at 100, 0.678 at 10^3 and 0.667 at 10^5. Within 5% of $\frac{2}{3}$ needs $\chi \gtrsim 200$. The classical end checks the calculation: at $\chi = 10^{-3}$ the power is 0.994095 of Larmor's, matching the known correction $1 - \frac{55\sqrt{3}}{16}\chi + 48\chi^2$. *Try it:* at fixed field, raising γ tenfold raises the power 100-fold classically but only $10^{2/3} = 4.64$-fold deep in the quantum regime.

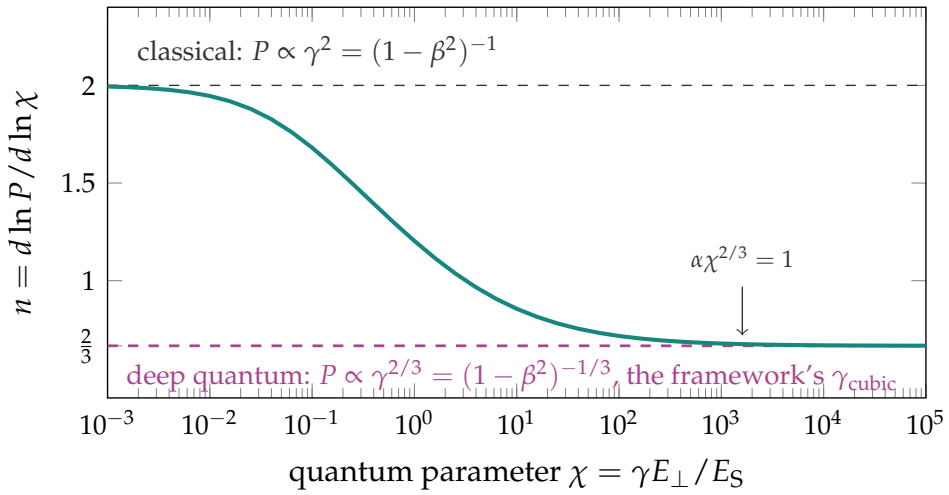


Figure 11: The power exponent of an electron radiating in a strong field slides from the classical 2 to the framework's $\frac{2}{3}$.

8.8 The cosmic jerk

The cosmological jerk $j = \dddot{a}/(aH^3)$ is the third time derivative of the scale factor, normalized by the Hubble time cubed; it is the statefinder parameter r of Sahni et al. [51]. For matter plus a cosmological constant it equals 1 at every epoch. The Engine maps a temporal volume $V_T \propto t^{n_c}$ to $w = n_c/2 - 1$, and its adopted static volume $n_c = 0$ is $w = -1$, so the framework predicts $j \equiv 1$. Its rejected alternative $n_c = 3$ (claim C5's $\rho_\Lambda \propto t^{-3}$) gives $j_0 = 3.36$.

Worked example 34: the jerk DESI's fits imply

For a flat universe with $w(a) = w_0 + w_a(1-a)$, $j = q(2q+1) - dq/d\ln a$. DESI DR2's best fits [32] give $j_0 = -1.41$ (DESI+CMB), -0.06 (+Pantheon+), -0.77 (+Union3) and -0.45 (+DESY5), each returning to 1.06–1.09 by $z = 1$ (Figure 12). The preference over $j \equiv 1$ is the paper's: 3.1, 2.8, 3.8 and 4.2σ. *Try it:* for constant w, $j = 1 + \frac{9}{2}w(1+w)\Omega_{\text{de}}$; any $w \neq -1$ moves j off 1.

8.9 Does a time dimension switch on?

The Engine writes the dark-energy density as $\rho_\Lambda \propto t^{-n_c}$, with n_c the number of the three time dimensions that carry the cosmic age. A count is an integer, and it can change. The smallest

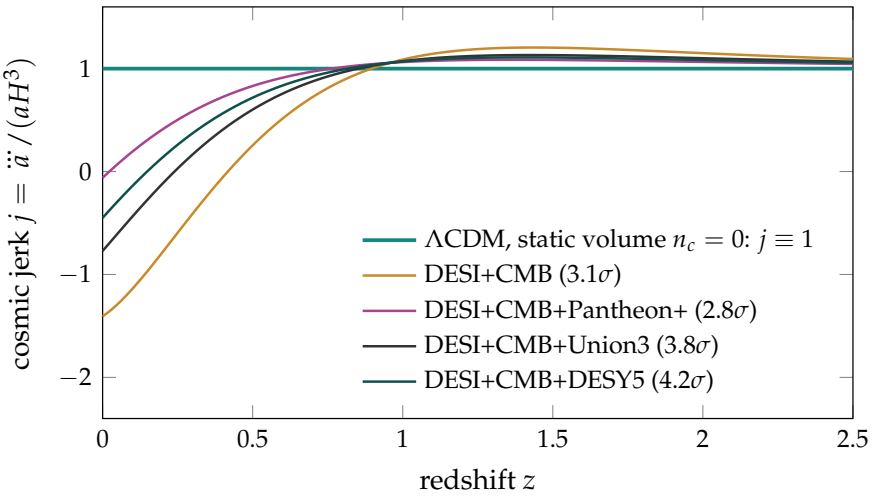


Figure 12: The cosmic jerk. A static temporal volume requires $j \equiv 1$; the DESI DR2 best fits do not.

change is $n_c : 0 \to 1$ at one cosmic time t_t, read literally:

$$\rho_{\rm DE}(t) = \begin{cases} \rho_e, & t < t_t, \\ \rho_e\, t_t/t, & t \geq t_t, \end{cases} \qquad \Longrightarrow \qquad w = -1 + \frac{1}{3Ht} \quad (t \geq t_t). \tag{26}$$

This is an extension of the framework, not something it states, and it has one parameter beyond ΛCDM: the switch-on redshift z_t. It was pre-registered before any fit, in a file (Appendix D) that fixed the data (DESI DR2 BAO, DESI's compressed CMB prior on $\theta_*, \omega_b, \omega_{bc}$, and the Pantheon+, Union3 and DES-SN5YR supernovae [32, 46, 47, 48]), a second reading as a secondary check (the Engine's matter-era mapping, $w = -\frac{1}{2}$ after z_t), and the verdict thresholds. Support required $\Delta\chi^2 \leq -6$ in every combination and a tie with DESI's two-parameter $w_0 w_a$ fit; even a partial result required $\Delta\chi^2 \leq -2$ in every combination. Distances come from CAMB [50]. The pipeline first had to reproduce DESI's own $w_0 w_a$ results, and it does. For DESI alone, with the compressed CMB, and with each supernova sample it finds $\Delta\chi^2 = -4.66, -7.91, -4.90, -10.07$ and -13.57, against DESI's $-4.7, -8.0, -4.9, -10.1$ and -13.6.

Worked example 35: the switch-on that the supernovae like best

DESI+CMB+DES-SN5YR prefers $z_t = 0.10$, with $\Omega_m = 0.3204$ and $H_0 = 66.19$. The switch happens at $t_t = 12.43$ Gyr, 1.35 Gyr ago. There $Ht = 0.914$, so w jumps from -1 to $-1 + 1/(3 \times 0.914) = -0.635$. Today $H_0 t_0 = 0.933$ and $w_0 = -0.643$; the dark-energy density has fallen by the factor $t_t/t_0 = 1/1.109$, or 10%, since the switch. *Try it:* in the matter era $Ht = \frac{2}{3}$, so eq. (26) gives $w = -\frac{1}{2}$, the Engine's $n_c = 1$.

data	$w_0 w_a$ $\Delta\chi^2$	switch-on $\Delta\chi^2$	best z_t	z_t at 3σ	matter-era reading
DESI+CMB	−7.91	−0.50	0.175	< 0.41	−0.49
+Pantheon+	−7.51	−3.18	0.040	< 0.11	−3.32
+Union3	−13.70	−8.59	0.150	< 0.26	−7.72
+DES-SN5YR	−17.89	−14.50	0.100	< 0.17	−14.62

Verdict: no support. With DESI and the CMB alone the switch-on gains only 0.5, below the pre-registered floor of 2, and in no combination does it tie $w_0 w_a$. The profiles (Figure 13) say three more things, labelled here as diagnostics rather than evidence.

- *An early switch-on is excluded.* $\Delta\chi^2$ climbs past $+100$ for $z_t \geq 1$ in every combination above. If one time dimension began to carry the cosmic age, it did so within the last 1.5–4.5 Gyr.
- *The preference is the supernovae's, and it moves with the sample.* The preferred onset is $z_t = 0.04$ (Pantheon+), 0.15 (Union3) or 0.10 (DES-SN5YR). The recalibrated DES sample [49] halves the gain, from -14.5 to -7.7, as it reduces w_0w_a's from -17.9 to -10.7. Without the CMB prior, a late switch-on fits DESI with the supernovae as well as w_0w_a does (-14.0 against -13.6 for DES-SN5YR, -9.5 against -10.1 for Union3): the supernova part of DESI's signal is a recent thaw, and a late step in w reproduces it.
- *What no count can do.* With the CMB, w_0w_a wins because its best fits cross $w = -1$ between $z = 0.39$ and 0.51 and are phantom ($w < -1$) beyond. A density $\propto t^{-n_c}$ with $n_c \geq 0$ never grows, so $w \geq -1$ always.

That leaves one clean prediction. Every temporal volume, static or switching, requires $w(z) \geq -1$ at every redshift. If DESI's later data or Euclid confirm a phantom crossing near $z \approx 0.4$–0.5, the whole family is excluded; if the crossing fades, the static volume, $n_c = 0$, stands.

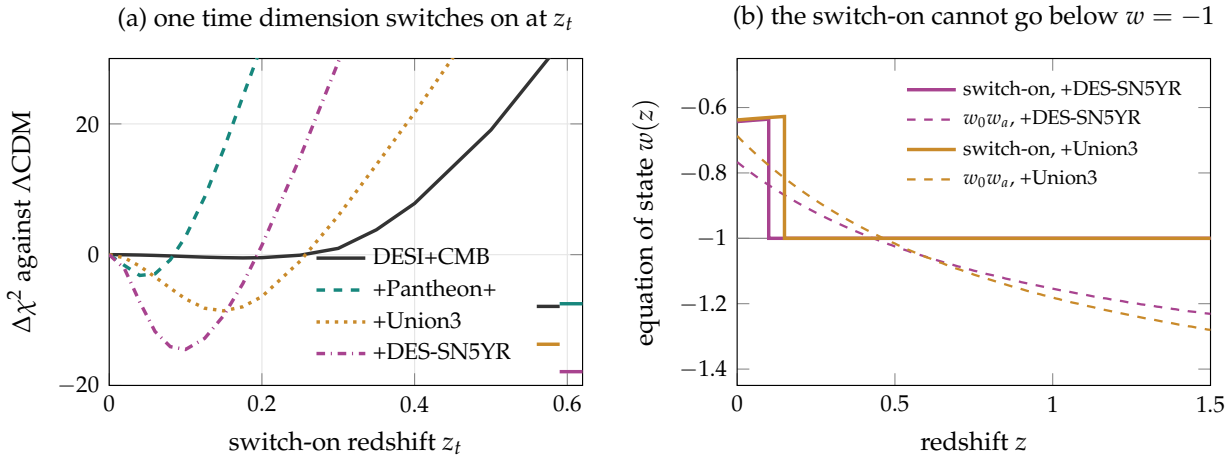


Figure 13: The switch-on test. (a) $\Delta\chi^2$ against ΛCDM as a function of the switch-on redshift; the short bars at the right edge are DESI's two-parameter w_0w_a fits to the same data. Above the plotted range the curves keep rising, past $+100$ by $z_t = 1$. (b) $w(z)$ for the best switch-ons (solid) and the w_0w_a best fits (dashed): the switch-on can only raise w above -1.

8.10 What to change in the framework

claim	status	replacement
C1: γ_{cubic} as a clock factor; "1,000 simulations"	excluded ($\lvert k-\frac{1}{2}\rvert < 1.9\times10^{-8}$ storage ring; $27\,000\sigma$ Galileo; 32σ NTS-2)	clocks keep the square root; the same exponent recurs, for unrelated reasons, in quantum-limit radiation, eq. (25)
C2: t_3 carries α; Scramble Zone	location right, mechanism wrong	$\tau_0 = \frac{2}{3}\alpha t_C$ carries α into dynamics; the zone is the radiation-reaction band, $Q \propto \alpha^{-3}$
C3: $\varphi = \omega t + \alpha t^3$	real on a bent path	coefficient $c^2/6\rho^2$, eq. (24); consequences γ^3, $K_{1/3}$, $K_{2/3}$, $\omega^{1/3}$
C4: normalized t^3–τ terms	no measurable content as written	the measured normalized cubic-time ratios: $\beta^3 = T_M/(P/2\pi)$, $Q/2\pi = \tau/T$, $j = \dddot{a}/(aH^3)$
C5: $\rho_\Lambda = \alpha_3/t^3$; $n_c = 0$	t^{-3} excluded; $n_c = 0$ under test; a switch $n_c : 0 \to 1$ not supported (§8.9)	$n_c = 0 \Leftrightarrow j \equiv 1$; DESI prefers $j_0 \neq 1$ at 2.8–4.2σ; every n_c gives $w \geq -1$, so a confirmed phantom crossing would exclude the family
$\alpha(t) = \alpha_0(1+\epsilon_2 t^2/t_0^2+\epsilon_3 t^3/t_0^3)$, $\epsilon_2 \sim 10^{-5}$, $\epsilon_3 \sim 10^{-6}$ (DE 82)	excluded: drifts at 1.7×10^{-15} yr^{-1} today, 1500σ from the clock bound [31]	$\lvert 2\epsilon_2 + 3\epsilon_3\rvert < 4.4\times10^{-8}$; the drift of such a law is largest today, not early
$\alpha = 24\pi^2\epsilon_0\hbar c/e^2$ (GP 16–19)	equals $6\pi/\alpha \approx 2583$	$\alpha = t_C/(a_0/c)$, $\alpha^3 = (r_e/c)/t_{\text{au}}$

Not established here. Equation (25) is a theorem of quantum electrodynamics; its asymptotic regime ($\chi \gtrsim 200$) has not been reached experimentally, and no experimental χ values are quoted because none were re-verified for this section. The DESI preference for $j_0 \neq 1$ is below 5σ. The double pulsar's intrinsic $\dot{P}_b$ was not located in the extracted text, so its comparison is with the observed value. The switch-on test uses DESI's compressed CMB prior, not the full Planck+ACT likelihood (with the full likelihood DESI's own w_0w_a gains are 3–4 larger), and its $\Delta\chi^2$ are best-fit differences; because z_t has no meaning under ΛCDM, they overstate significance, which only strengthens the negative verdict. The $\alpha(t)$ row uses the paper's own coefficients; a smaller pair would pass the clocks and predict no measurable variation.

9 The first edition's fitted times on the clock ladder

For hydrogen the derived clocks form a ladder whose rungs are set by successive powers of $Z\alpha$ (Figure 14):

- the Compton clock, $\hbar/mc^2 = 1.288\times10^{-21}$ s (α^0);
- the binding beat, $\hbar/B = 4.84\times10^{-17}$ s, and the orbital period, 1.52×10^{-16} s (both α^{-2}; the orbit is a full period, the others are $\hbar/E$ times);
- the fine-structure time, 1.45×10^{-11} s (α^{-4});
- the 2S–2P Lamb-shift time, 1.5×10^{-10} s.

The first edition's Initial $t_1 = 1.29\times10^{-21}$ s is the Compton clock. Its final t_1 and t_2 sit at round ratios to it, $t_1/t_C = 1/7.00$ and $t_2/t_C = 10.01$. This may be the fit rediscovering the proper clock

through a rescaling, or a coincidence at three-digit precision [OPEN]. The final $t_3 = 5.35 \times 10^{-13}$ s matches no rung; it falls between the binding beat and the fine-structure time.

Worked example 36: climbing hydrogen's clock ladder

With $\hbar = 6.582119569 \times 10^{-16}$ eV s: *Compton:* $\hbar/mc^2 = 1.288089 \times 10^{-21}$ s. *Binding beat:* $\hbar/B = 6.582119569 \times 10^{-16}/13.598468 = 4.8403 \times 10^{-17}$ s. At leading order $B = \alpha^2 mc^2/2$, so $\hbar/B = 2t_C/\alpha^2 = 4.838 \times 10^{-17}$ s, two powers of α^{-1} above the Compton clock. *Orbital period:* $2\pi t_{\text{au}} = 2\pi \times 2.418884 \times 10^{-17} = 1.5198 \times 10^{-16}$ s. *Fine structure:* $\hbar/\Delta E = 6.582119569 \times 10^{-16}/4.525951 \times 10^{-5} = 1.4543 \times 10^{-11}$ s. At leading order $\Delta E = \alpha^4 mc^2/32$, so $\hbar/\Delta E = 32\, t_C/\alpha^4 = 1.4536 \times 10^{-11}$ s. *Lamb:* $1/(2\pi \times 1057.8\ \text{MHz}) = 1.5046 \times 10^{-10}$ s. *The first edition's final times:* $t_1/t_C = 1.84 \times 10^{-22}/1.288089 \times 10^{-21} = 0.14285 = 1/7.000$ and $t_2/t_C = 1.29 \times 10^{-20}/1.288089 \times 10^{-21} = 10.015$. *Try it:* for He^+ the Compton rung stays where it is, the binding beat falls by $Z^2 = 4$ to $\hbar/54.41821$ eV $= 1.21 \times 10^{-17}$ s, and the fine-structure time falls by $Z^4 = 16$.

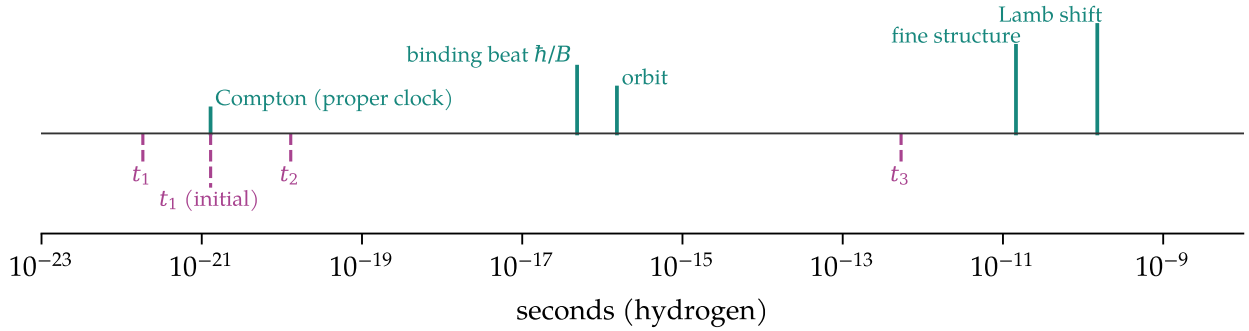


Figure 14: Hydrogen's derived clock ladder (teal) and the first edition's fitted times (magenta, dashed).

10 Spin in the cubic-time framework (retained, with corrections)

The first edition's spin sector (1st-ed. eq. (7)–(23)) is the part of the framework with dynamics of its own, and it is retained here in summary. The spin state evolves as $\chi(t_1, t_2, t_3) = \chi_0 \exp[i\omega_s(t_1 + t_2 + t_3)]$, where ω_s is the spin frequency of 1st-ed. eq. (7). The spinor of 1st-ed. eq. (19) carries the orientation θ and phase ϕ, and the Lagrangian

$$L = \tfrac{1}{2} m(t_1, t_2, t_3)\, \dot{x}^2 + \tfrac{1}{2} I(\theta)\, \dot{\theta}^2 + \lambda \cos\phi\, \dot{\theta} \qquad \text{(1st ed. 20)}$$

couples mass-energy to spin. Three corrections apply.

S1. 1st-ed. eq. (9) and (23). $\omega_p = \frac{\omega_s}{2}(t_1 - t_2)$ is dimensionally a phase, not a frequency. The accumulated angle is $\Delta\phi_p = \frac{\omega_s}{2}(t_1 - t_2)$, and the precession frequency is $\omega_p = \frac{\omega_s}{2}\frac{d}{dt}(t_1 - t_2)$. Field-free precession then requires the two time coordinates to advance at different rates. A clock-rate difference already produces a precession in standard physics, Thomas precession, so any prediction must be distinguished from it. It also needs a preferred axis, which ω_s alone does not supply [OPEN].

S2. 1st-ed. eq. (11)–(13). $\hat{S}_{t_i} = \hbar\omega_s\, \partial/\partial t_i$ has units of energy per time. An angular-momentum operator needs $\hat{S}_{t_i} = -i\hbar\, \partial/\partial(\omega_s t_i)$, whose eigenvalue on $e^{i\omega_s t_i}$ is $\hbar$.

S3. 1st-ed. eq. (22). L contains no $\dot{\phi}$, so the Euler–Lagrange equation for ϕ reduces to the constraint $\partial L/\partial\phi = -\lambda \sin\phi\, \dot{\theta} = 0$. As written, this freezes ϕ at 0 or π, or stops θ. A kinetic term $\frac{1}{2} J\dot{\phi}^2$ is needed for the phase to have dynamics [OPEN].

Worked example 37: Thomas precession of hydrogen's 1s electron, and the dimensions of S2

Thomas precession (S1). A spin carried around a circle at speed βc precesses at $\omega_T = (\gamma - 1)\,\omega$ with no field at all. For the 1s electron $\beta = \alpha$, so $\gamma - 1 = 1/\sqrt{1-\alpha^2} - 1 = 2.662674 \times 10^{-5}$. The orbital angular frequency is $\omega = \alpha c/a_0 = 0.0072973526 \times 299\,792\,458/5.29177 \times 10^{-11} = 4.1341 \times 10^{16}$ rad/s, so $\omega_T = 1.1008 \times 10^{12}$ rad/s, or $2\pi(\gamma - 1) = 1.6730 \times 10^{-4}$ rad per orbit. Any field-free precession predicted by the corrected eq. (9) has to be told apart from this. *Dimensions (S2).* On $e^{i\omega_s t_i}$, the corrected operator $-i\hbar\,\partial/\partial(\omega_s t_i)$ returns $-i\hbar \cdot i = \hbar$, an angular momentum; the first edition's $\hbar\omega_s\,\partial/\partial t_i$ returns $i\hbar\omega_s^2$, in J/s. *Try it:* for the 1s electron of U^{91+}, $\beta = 0.6714$ and $\gamma - 1 = 0.3493$, so the spin turns 2.19 rad per orbit.

The "human body" application of the temporal-volume conversion, 1st-ed. eq. (24) ($V_{\text{temporal}} = V_{\text{spatial}}/t_1 t_2$, with units of m^3/s^2), lies outside the scope of this atomic paper and is not carried forward.

11 Discussion

What is new and what is not. The Dirac virial theorem, the Dirac–Coulomb spectrum, mean-field atoms and the Gamow factor are standard physics; every number here could be computed without the temporal reading. The premise itself has a long pedigree:

- its classical form, that a bound system's energy is the time average of $\sum mc^2\sqrt{1 - v^2/c^2}$, is Landau and Lifshitz's relativistic virial theorem [3, 4, 5];
- its Dirac form is Fock's [2] and Rose and Welton's [1], and its Hellmann–Feynman derivation is Adkins's [6];
- the many-electron form is Bahcall's [8]; Matsuoka and Koga added the finite nucleus [9], and Froese Fischer and Godefroid the orbital relations [10].

What this paper adds is the reading of those theorems as clock rates, carried through all 118 elements with the identity closed numerically, and the split of each orbital's non-Coulomb term into shelter and exchange. That is a contribution of organization and interpretation, not a new law. The framework's labelling earns its place in four ways.

- It states exactly where the premise holds and where it fails.
- It turns screening into a ledger of clock, shelter and exchange terms.
- It exposes a literal cube of time in the radiative decay law.
- It connects the atom's clock ($\beta = Z\alpha/n$) to the alpha-decay clock ($\eta = Z_1 Z_2 \alpha/\beta$) through a single ratio.

What lies beyond the construction. Each of the following remains visible as a residual:

- the Lamb shift, from QED and the finite nuclear size;
- the Breit interaction;
- correlation beyond the local-density model, the largest valence errors;
- core polarization, in the alkali lifetimes.

The spin sector's field-free precession is the framework's one proposed dynamical prediction that no spatial result supplies. Stating it with correct dimensions (S1) and separating it from Thomas precession is the next step.

The constant α and t_3. The first edition proposed that t_3 sets the fine-structure coupling. The sensitivity of a line to α is $K = d\ln\omega/d\ln\alpha$. In the non-relativistic atom every line has $K = 2$, which cannot be distinguished from a redshift or a change of units, so only the relativistic, proper-time part $K - 2$ is observable. For hydrogen's Lyman-α that part is 3.3×10^{-5}; for U^{91+} it is 0.43. Atomic clocks measure $\dot{\alpha}/\alpha = (1.0 \pm 1.1) \times 10^{-18}$ yr^{-1} [31]. Suppose one of the time dimensions tracked cosmic age and α depended on it, with $S = d\ln\alpha/d\ln t_i$. Then α would drift at S/t_0, with $t_0 = 13.8$ Gyr, and the clock bound requires $|S| < 4.4 \times 10^{-8}$ at 2σ. The same experiment bounds α's response to gravitational potential, $(c^2/\alpha)\, d\alpha/d\Phi = (14 \pm 11) \times 10^{-9}$ [31]. The S bound is worked in Section 7.4.

Worked example 38: the sensitivity of a line to α

Hydrogen's Lyman-α (1s–2p$_{3/2}$, infinitely heavy nucleus) is $\hbar\omega = mc^2[\langle d\tau/dt\rangle_{2p_{3/2}} - \langle d\tau/dt\rangle_{1s}]$. Expanding $\sqrt{1-\alpha^2/4}$ and $\sqrt{1-\alpha^2}$, $\hbar\omega = mc^2\left[\frac{3}{8}\alpha^2 + \frac{15}{128}\alpha^4\right]$: the α^2 term is Bohr's, and the α^4 term is the relativistic part of the two clocks. Then $K = d\ln\omega/d\ln\alpha = 2 + 2\alpha^2\,(15/128)/(3/8) = 2 + \frac{5}{8}\alpha^2$, so $K - 2 = \frac{5}{8} \times 5.3251 \times 10^{-5} = 3.33 \times 10^{-5}$. Differentiating the exact rates numerically gives 3.3283×10^{-5}. For U^{91+}'s Lyα_1, where relativity is not small, the same derivative gives $K = 2.4324$, so $K - 2 = 0.432$. *Why only $K - 2$ is observable:* a change of α that rescaled every line by the same α^2 could be absorbed into the unit of energy; only the part of K that differs from line to line, the proper-time part, can be seen. *Try it:* He^+'s Lyman-α has $x = 4\alpha^2$, so $K - 2 \approx \frac{5}{8} \times 4\alpha^2 = 1.33 \times 10^{-4}$.

12 Conclusion

The first edition asked whether an electron's states could be found with time. With two clocks and no per-atom fitting, they can: exactly for one-electron atoms, and for every element within a mean-field model. Locking the electron's clock to the state's clock gives $E = mc^2\langle d\tau/dt\rangle$, exact for the inverse-square force. It reproduces hydrogen, deuterium and He^+ to the Lamb shift and the X-ray lines to 1%. Summed over electrons, it gives the total energy of every mean-field atom in the periodic table, as an identity closed to 10^{-7} once the nuclear-volume term is included. With shelter and exchange added, it gives ionization energies to 5% and K-shell binding energies to 0.5% for $Z \geq 10$. The same temporal language describes how atomic states end: by a cube of the light's period, and by the Sommerfeld count at the nuclear barrier. Read as a test of the wider temporal framework, the atom confirms its structural claims and measures its dilation exponent independently: $k = 0.49996 \pm 0.00010$, consistent with the storage-ring bound $|k - \frac{1}{2}| < 1.9 \times 10^{-8}$; clocks follow the square root of special relativity, not the cube root. Cubic time itself contributes, but to radiation rather than to clocks: as the jerk of radiation reaction, as mass weighed by an orbit, and as the arrival time of light from a bent path (Section 8). The first edition's intuition that time can carry the atom was right; what it needed was the second clock.

A Numerical methods and checks

- **Integration.** The radial Dirac equations for $P = rg$ and $Q = rf$ are solved on a logarithmic grid of 3200 points from 10^{-7} to 60 bohr, by fourth-order Runge–Kutta shooting. Midpoint

potentials use four-point Lagrange interpolation of rV. Energies come from node-count bisection with the matching correction $\delta E = cP(r_m)[Q_{\text{out}} - Q_{\text{in}}]/\int(P^2 + Q^2)$, converged to 10^{-11}.

- **Solver validation.** Against the exact Dirac–Coulomb levels, the error is below 1.3×10^{-10} for $Z \leq 26$ and below 5×10^{-8} at $Z = 92$. The finite-nucleus 1s shift in U is 193.8 eV.
- **Self-consistency.** The potential is linearly mixed until its maximum relative change falls below 10^{-9}. All 118 atoms and all transition states converged.
- **Identity checks.** The whole-atom ledger closes to 1.1×10^{-7} in the exchange-only atoms. The per-orbital identity (8) closes to 0.05 eV absolute in orbitals bound by up to 220 keV; the residual comes from the numerical derivative of rV.
- **Nuclear model.** A uniform sphere with $R = \sqrt{5/3}\, r_{\text{rms}}$. Here $r_{\text{rms}} = 0.8414$ fm for $A = 1$, $1.0 + 0.3A^{1/3}$ fm for $A < 10$, and $0.836A^{1/3} + 0.570$ fm otherwise. A is the mass number of the standard atomic weight, or of the longest-lived isotope.
- **Alkali lifetimes.** The ground and excited configurations are each solved self-consistently (spin-polarized), and the radial dipole is taken between them.
- **Alpha decay.** Q values are taken from AME2020 atomic masses [29], cross-checked against a second tabulation. Half-lives are alpha partial half-lives where the branching is below 99.9%.

B Data sources

- **First ionization energies:** NIST Atomic Spectra Database [36] for $Z \leq 108$, and the element pages of Wikipedia for the theoretical estimates at $Z = 109$–118.
- **K-shell binding and Kα lines:** X-ray Data Booklet, Tables 1-1 and 1-2 [26].
- **Transition probabilities:** NIST Handbook of Basic Atomic Spectroscopic Data [27]. Hydrogen 2p: NIST ASD.
- **Half-lives:** NUBASE2020 [28] as quoted on the Wikipedia isotope pages, which also give the newer values used for Sm-146 (2024) and Gd-148.
- **1S Lamb shifts:** Weitz et al. [19].
- **U^{91+}:** Gumberidze et al. [22].
- **Deuterium Lyman-α:** Herzberg [20], with the component analysis of [21].
- **One-electron ions, theory and experiment (Section 7.1):** Yerokhin and Shabaev [23], Table V.
- **Alpha-decay Q values:** AME2020 [29].
- **Constants:** CODATA 2022 [24] (Appendix C).
- **Cubic-time tests (Section 8):** Galileo 5 and 6 [37]; GPS and NTS-2 [38]; PSR B1913+16 [39] and J0737−3039A/B [40]; DESI DR2 [32].

C Constants used

Every worked example uses the CODATA 2022 values below [24]; derived quantities are given with the formula that produces them. Substituting these into the boxed examples reproduces their digits; the few steps that need a numerical solution (helium, the finite nucleus) name the script that supplies them.

quantity	symbol	value
fine-structure constant	α	$7.2973525643 \times 10^{-3}$ ($1/\alpha = 137.035999177$)
electron rest energy	mc^2	510 998.95069 eV
Rydberg energy	$R_\infty hc$	13.605693122990 eV
Rydberg constant	R_∞	10 973 731.568157 m^{-1}
Hartree energy	E_h	27.211386245981 eV
reduced Planck constant	$\hbar$	$6.582119569 \times 10^{-16}$ eV s $= 1.054571817 \times 10^{-34}$ J s
Planck constant	h, hc	$4.135667696 \times 10^{-15}$ eV s; 1239.841984 eV nm
speed of light	c	299 792 458 m/s (exact)
Bohr radius	a_0	$5.29177210544 \times 10^{-11}$ m
Compton time	$t_C = \hbar/mc^2$	1.288089×10^{-21} s
atomic unit of time	$t_{\mathrm{au}} = \hbar/E_h$	2.418884×10^{-17} s
mass ratios	m_p/m_e; m_d/m_e; m_α/m_e	1836.152673426; 3670.482967655; 7294.29954171
reduced-mass factors	$\mu/m = (1 + m_e/M)^{-1}$	H 0.9994556794; D 0.9997276305; He^+ 0.9998629254
atomic mass unit	$\mathrm{u}\,c^2$	931.49410372 MeV
Coulomb constant	$e^2/4\pi\varepsilon_0 = \alpha\hbar c$	1.439965 MeV fm
Newtonian constant	G	6.67430×10^{-11} $\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$
proton mass	m_p	$1.67262192595 \times 10^{-27}$ kg
Earth's gravitational parameter	$GM_\oplus$	$3.986004418 \times 10^{14}$ $\mathrm{m}^3/\mathrm{s}^2$ (IERS)

D Reproducing the numbers

The worked examples, Table 4, Figures 7 and 8 and the erratum of Appendix F come from the Python scripts distributed with this revision (Python 3 with NumPy and SciPy), using the data files beside them. The 118-element calculations and Figures 1–10 are those of the second edition. Each script prints its numbers in the order of the examples it serves; run it with `python3` from the `scripts/` directory.

script	what it computes, and where it is used
`chrono_constants.py`	the constants of Appendix C
`we_onee.py`	the premise, the clock rates and the binding energies (Section 3); Table 1 row by row; the clock law and Table 4 (Section 7.1)
`we_dirac_numeric.py`	a radial Dirac solver: $\langle\beta\rangle$ against E/mc^2, the Hellmann–Feynman check, and the finite nucleus of U^{91+} (Worked example 11)
`we_helium.py`	numerical Hartree–Fock helium and the screened hand model (the clock–shelter ledger)

script	what it computes, and where it is used
`we_decay.py`	the cube law for H, He^+ and the alkalis; the Gamow factor for U-238 and Po-212; all 35 alpha emitters recomputed from AME2020 (Appendix F)
`we_xray.py`	molybdenum's Kα lines; the RMS errors over all 80 Kα_1 lines, and the doublet for $Z \geq 40$
`we_framework.py`	the audit numbers (Section 2); the two kinds of slow clock, α as a ratio of clocks and the S bound (Sections 7.2–7.4); $K - 2$; Thomas precession; the clock ladder
`make_figures.py`	the pgfplots sources of Figures 7 and 8
data file	**source**
`mass_1.mas20.txt`	AME2020 atomic masses [29], read by `we_decay.py`
`xdb_sec1_2.txt`	X-ray Data Booklet, Table 1-2 [26], read by `we_xray.py`
`codata_allascii.txt`	CODATA 2022 from NIST [24], the reference for the values in `chrono_constants.py`

To rebuild the two new figures:

```
in scripts/:   python3 make_figures.py
in src/:       tectonic -X compile f11_clocklaw.tex
               tectonic -X compile f12_twoclocks.tex
```

Section 8. The cubic-time results come from the scripts in `scripts_cubic_time/` (run there with `python3`); each literature value, with how it was read, is in `literature_values.md` in the same folder.

script	content
`ct_gnss.py`	Section 8.2: GPS offset, NTS-2, Galileo 5/6 under eq. (18)
`ct_kepler.py`	Section 8.3: mass as cubic time; exact orbital clock
`ct_pulsar.py`	Section 8.4: post-Keplerian parameters; masses and $\dot{P}_b$
`ct_radiation.py`	Section 8.5: τ_0, the jerk drag, collapse time, $A/\tau_0\omega^2$, $Q\alpha^3$
`ct_synchrotron.py`	Section 8.6: retarded-field integration against the Bessel form
`ct_quantum.py`	Section 8.7: quantum radiated power and its exponent
`ct_cosmo.py`	Section 8.8: the cosmic jerk for ΛCDM, n_c and DESI
`ct_figures.py`	Figures 9–12 (pgfplots)
`ct_alpha_drift.py`	Section 8.10: the $\alpha(t)$ law's drift against the clock bound
`newphysics/desi_nc.py`	Section 8.9: the switch-on test, its validation against DESI, the profiles; `nc_summary.py` prints the table, `nc_figure.py` writes Figure 13; the hypothesis and thresholds were fixed first in `PREREGISTRATION.md` (hashed), the outcome is in `RESULTS.md`

E The periodic table in proper time

Columns.

- Configuration: NIST ground state; * marks a prediction.
- IE: first ionization energy in eV, computed and reference. Marks: † theoretical estimate; ‡ NIST value from QED theory anchored to spectroscopy.
- K: K-shell binding, computed and X-ray Data Booklet, in eV (k = keV); errors for $Z \geq 10$. For H and He the K-shell binding is the first ionization energy.

- Clock: proper time lost per day by the 1s electron and by the electron that ionizes. For hydrogen the model's value, 2.15 s, carries the local-density self-interaction error; the exact value is 2.30 s.

Z	El.	configuration	IE calc	IE ref	err %	K calc	K meas	err %	1s clock	valence
1	H	$1s^1$	13.351	13.5984‡	−1.8	13.4	13.6	–	2.15 s	2.15 s
2	He	$1s^2$	24.632	24.5874	+0.2	24.6	24.6	–	6.36 s	6.36 s
3	Li	[He] $2s^1$	5.584	5.3917	+3.6	66.9	54.7	–	16.11 s	1.14 s
4	Be	[He] $2s^2$	9.101	9.3227	−2.4	126.5	111.5	–	30.31 s	2.61 s
5	B	[He] $2s^2\,2p^1$	8.458	8.2980	+1.9	204.4	188.0	–	49.07 s	3.52 s
6	C	[He] $2s^2\,2p^2$	11.599	11.2603	+3.0	300.6	284.2	–	72.34 s	5.90 s
7	N	[He] $2s^2\,2p^3$	14.804	14.5341	+1.9	415.2	409.9	–	100.13 s	8.81 s
8	O	[He] $2s^2\,2p^4$	13.878	13.6181	+1.9	550.0	543.1	–	2.2 min	11.85 s
9	F	[He] $2s^2\,2p^5$	17.972	17.4228	+3.2	703.1	696.7	–	2.8 min	15.54 s
10	Ne	[He] $2s^2\,2p^6$	22.036	21.5645	+2.2	874.7	870.2	+0.51	3.5 min	19.81 s
11	Na	[Ne] $3s^1$	5.439	5.1391	+5.8	1.084 k	1.071 k	+1.19	4.3 min	1.72 s
12	Mg	[Ne] $3s^2$	7.776	7.6462	+1.7	1.316 k	1.303 k	+0.98	5.1 min	3.15 s
13	Al	[Ne] $3s^2\,3p^1$	5.934	5.9858	−0.9	1.573 k	1.560 k	+0.86	6.0 min	3.05 s
14	Si	[Ne] $3s^2\,3p^2$	8.180	8.1517	+0.4	1.854 k	1.839 k	+0.79	7.0 min	4.84 s
15	P	[Ne] $3s^2\,3p^3$	10.430	10.4867	−0.5	2.157 k	2.146 k	+0.55	8.1 min	6.88 s
16	S	[Ne] $3s^2\,3p^4$	10.559	10.3600	+1.9	2.485 k	2.472 k	+0.54	9.3 min	8.98 s
17	Cl	[Ne] $3s^2\,3p^5$	13.222	12.9676	+2.0	2.837 k	2.822 k	+0.50	10.5 min	11.39 s
18	Ar	[Ne] $3s^2\,3p^6$	15.873	15.7596	+0.7	3.212 k	3.206 k	+0.18	11.8 min	14.08 s
19	K	[Ar] $4s^1$	4.586	4.3407	+5.7	3.620 k	3.608 k	+0.32	13.2 min	1.86 s
20	Ca	[Ar] $4s^2$	6.263	6.1132	+2.4	4.054 k	4.038 k	+0.37	14.6 min	3.05 s
21	Sc	[Ar] $3d^1\,4s^2$	6.684	6.5615	+1.9	4.506 k	4.492 k	+0.32	16.2 min	3.57 s
22	Ti	[Ar] $3d^2\,4s^2$	6.979	6.8281	+2.2	4.983 k	4.966 k	+0.35	17.8 min	3.97 s
23	V	[Ar] $3d^3\,4s^2$	7.221	6.7462	+7.0	5.484 k	5.465 k	+0.36	19.5 min	4.31 s
24	Cr	[Ar] $3d^5\,4s^1$	7.585	6.7665	+12.1	6.004 k	5.989 k	+0.25	21.3 min	4.31 s
25	Mn	[Ar] $3d^5\,4s^2$	7.627	7.4340	+2.6	6.560 k	6.539 k	+0.32	23.2 min	4.91 s
26	Fe	[Ar] $3d^6\,4s^2$	8.204	7.9025	+3.8	7.135 k	7.112 k	+0.32	25.1 min	5.33 s
27	Co	[Ar] $3d^7\,4s^2$	8.692	7.8810	+10.3	7.735 k	7.709 k	+0.34	27.1 min	5.71 s
28	Ni	[Ar] $3d^8\,4s^2$	9.130	7.6399	+19.5	8.360 k	8.333 k	+0.32	29.2 min	6.05 s
29	Cu	[Ar] $3d^{10}\,4s^1$	8.596	7.7264	+11.3	9.003 k	8.979 k	+0.27	31.4 min	5.37 s
30	Zn	[Ar] $3d^{10}\,4s^2$	9.921	9.3942	+5.6	9.685 k	9.659 k	+0.26	33.7 min	6.70 s
31	Ga	[Ar] $3d^{10}\,4s^2\,4p^1$	5.972	5.9993	−0.5	10.395 k	10.367 k	+0.27	36.0 min	4.27 s
32	Ge	[Ar] $3d^{10}\,4s^2\,4p^2$	7.963	7.8994	+0.8	11.133 k	11.103 k	+0.27	38.5 min	6.21 s
33	As	[Ar] $3d^{10}\,4s^2\,4p^3$	9.872	9.7886	+0.9	11.897 k	11.867 k	+0.25	41.0 min	8.29 s
34	Se	[Ar] $3d^{10}\,4s^2\,4p^4$	9.930	9.7524	+1.8	12.689 k	12.658 k	+0.25	43.6 min	10.37 s
35	Br	[Ar] $3d^{10}\,4s^2\,4p^5$	12.109	11.8138	+2.5	13.509 k	13.474 k	+0.26	46.3 min	12.64 s
36	Kr	[Ar] $3d^{10}\,4s^2\,4p^6$	14.220	13.9996	+1.6	14.356 k	14.326 k	+0.21	49.0 min	15.07 s
37	Rb	[Kr] $5s^1$	4.431	4.1771	+6.1	15.237 k	15.200 k	+0.24	51.9 min	2.29 s
38	Sr	[Kr] $5s^2$	5.889	5.6949	+3.4	16.147 k	16.105 k	+0.26	54.8 min	3.54 s
39	Y	[Kr] $4d^1\,5s^2$	6.400	6.2173	+2.9	17.083 k	17.038 k	+0.26	57.8 min	9.96 s
40	Zr	[Kr] $4d^2\,5s^2$	6.767	6.6341	+2.0	18.047 k	17.998 k	+0.27	61.0 min	4.89 s
41	Nb	[Kr] $4d^4\,5s^1$	7.385	6.7588	+9.3	19.035 k	18.986 k	+0.26	64.2 min	5.18 s
42	Mo	[Kr] $4d^5\,5s^1$	7.699	7.0924	+8.6	20.055 k	20.000 k	+0.27	67.4 min	5.62 s
43	Tc	[Kr] $4d^5\,5s^2$	7.349	7.1194	+3.2	21.108 k	21.044 k	+0.30	70.8 min	6.11 s
44	Ru	[Kr] $4d^7\,5s^1$	7.993	7.3605	+8.6	22.182 k	22.117 k	+0.29	74.3 min	6.14 s
45	Rh	[Kr] $4d^8\,5s^1$	8.087	7.4589	+8.4	23.289 k	23.220 k	+0.30	77.8 min	6.32 s
46	Pd	[Kr] $4d^{10}$	9.148	8.3368	+9.7	24.422 k	24.350 k	+0.30	81.5 min	29.14 s
47	Ag	[Kr] $4d^{10}\,5s^1$	8.229	7.5762	+8.6	25.593 k	25.514 k	+0.31	85.2 min	6.58 s
48	Cd	[Kr] $4d^{10}\,5s^2$	9.416	8.9938	+4.7	26.794 k	26.711 k	+0.31	89.0 min	8.00 s
49	In	[Kr] $4d^{10}\,5s^2\,5p^1$	5.681	5.7864	−1.8	28.029 k	27.940 k	+0.32	93.0 min	4.75 s
50	Sn	[Kr] $4d^{10}\,5s^2\,5p^2$	7.391	7.3439	+0.6	29.294 k	29.200 k	+0.32	97.0 min	6.56 s
51	Sb	[Kr] $4d^{10}\,5s^2\,5p^3$	8.998	8.6084	+4.5	30.592 k	30.491 k	+0.33	101.1 min	8.45 s

Z	El.	configuration	IE calc	IE ref	err %	K calc	K meas	err %	1s clock	valence
52	Te	[Kr] $4d^{10}\,5s^2\,5p^4$	9.078	9.0098	+0.8	31.921 k	31.814 k	+0.34	105.3 min	10.32 s
53	I	[Kr] $4d^{10}\,5s^2\,5p^5$	10.857	10.4512	+3.9	33.283 k	33.169 k	+0.34	109.6 min	12.30 s
54	Xe	[Kr] $4d^{10}\,5s^2\,5p^6$	12.560	12.1298	+3.5	34.676 k	34.561 k	+0.33	114.0 min	14.38 s
55	Cs	[Xe] $6s^1$	4.129	3.8939	+6.0	36.109 k	35.985 k	+0.34	118.5 min	2.53 s
56	Ba	[Xe] $6s^2$	5.395	5.2117	+3.5	37.574 k	37.441 k	+0.36	2.05 h	3.76 s
57	La	[Xe] $5d^1\,6s^2$	5.838	5.5769	+4.7	39.071 k	38.925 k	+0.38	2.13 h	4.50 s
58	Ce	[Xe] $4f^1\,5d^1\,6s^2$	5.906	5.5386	+6.6	40.594 k	40.443 k	+0.37	2.21 h	4.61 s
59	Pr	[Xe] $4f^3\,6s^2$	5.645	5.4702†	+3.2	42.144 k	41.991 k	+0.36	2.29 h	4.16 s
60	Nd	[Xe] $4f^4\,6s^2$	5.698	5.5248	+3.1	43.734 k	43.569 k	+0.38	2.37 h	4.24 s
61	Pm	[Xe] $4f^5\,6s^2$	5.750	5.5819	+3.0	45.359 k	45.184 k	+0.39	2.46 h	4.32 s
62	Sm	[Xe] $4f^6\,6s^2$	5.802	5.6437	+2.8	47.019 k	46.834 k	+0.39	2.55 h	4.40 s
63	Eu	[Xe] $4f^7\,6s^2$	5.854	5.6704	+3.2	48.713 k	48.519 k	+0.40	2.63 h	4.47 s
64	Gd	[Xe] $4f^7\,5d^1\,6s^2$	6.289	6.1498	+2.3	50.451 k	50.239 k	+0.42	2.72 h	5.21 s
65	Tb	[Xe] $4f^9\,6s^2$	6.099	5.8638	+4.0	52.212 k	51.996 k	+0.41	2.82 h	4.69 s
66	Dy	[Xe] $4f^{10}\,6s^2$	6.201	5.9391	+4.4	54.016 k	53.789 k	+0.42	2.91 h	4.78 s
67	Ho	[Xe] $4f^{11}\,6s^2$	6.298	6.0215	+4.6	55.857 k	55.618 k	+0.43	3.01 h	4.87 s
68	Er	[Xe] $4f^{12}\,6s^2$	6.392	6.1077	+4.7	57.736 k	57.486 k	+0.43	3.10 h	4.96 s
69	Tm	[Xe] $4f^{13}\,6s^2$	6.484	6.1844	+4.9	59.653 k	59.390 k	+0.44	3.20 h	5.05 s
70	Yb	[Xe] $4f^{14}\,6s^2$	6.576	6.2542	+5.2	61.609 k	61.332 k	+0.45	3.31 h	5.15 s
71	Lu	[Xe] $4f^{14}\,5d^1\,6s^2$	5.502	5.4259	+1.4	63.611 k	63.314 k	+0.47	3.41 h	10.60 s
72	Hf	[Xe] $4f^{14}\,5d^2\,6s^2$	6.959	6.8251	+2.0	65.655 k	65.351 k	+0.47	3.52 h	14.02 s
73	Ta	[Xe] $4f^{14}\,5d^3\,6s^2$	7.788	7.5496	+3.2	67.740 k	67.416 k	+0.48	3.62 h	7.60 s
74	W	[Xe] $4f^{14}\,5d^4\,6s^2$	8.005	7.8640	+1.8	69.867 k	69.525 k	+0.49	3.73 h	8.21 s
75	Re	[Xe] $4f^{14}\,5d^5\,6s^2$	8.186	7.8335	+4.5	72.037 k	71.676 k	+0.50	3.84 h	8.78 s
76	Os	[Xe] $4f^{14}\,5d^6\,6s^2$	8.911	8.4382	+5.6	74.250 k	73.871 k	+0.51	3.96 h	9.53 s
77	Ir	[Xe] $4f^{14}\,5d^7\,6s^2$	9.493	8.9670	+5.9	76.507 k	76.111 k	+0.52	4.07 h	10.20 s
78	Pt	[Xe] $4f^{14}\,5d^9\,6s^1$	9.776	8.9588	+9.1	78.805 k	78.395 k	+0.52	4.19 h	10.04 s
79	Au	[Xe] $4f^{14}\,5d^{10}\,6s^1$	9.890	9.2256	+7.2	81.152 k	80.725 k	+0.53	4.31 h	10.36 s
80	Hg	[Xe] $4f^{14}\,5d^{10}\,6s^2$	10.865	10.4375	+4.1	83.549 k	83.102 k	+0.54	4.44 h	11.93 s
81	Tl	[Xe] $4f^{14}\,5d^{10}\,6s^2\,6p^1$	6.176	6.1083	+1.1	85.995 k	85.530 k	+0.54	4.56 h	7.24 s
82	Pb	[Xe] $4f^{14}\,5d^{10}\,6s^2\,6p^2$	7.599	7.4167	+2.5	88.490 k	88.005 k	+0.55	4.69 h	9.33 s
83	Bi	[Xe] $4f^{14}\,5d^{10}\,6s^2\,6p^3$	7.103	7.2855	−2.5	91.035 k	90.524 k	+0.56	4.82 h	8.49 s
84	Po	[Xe] $4f^{14}\,5d^{10}\,6s^2\,6p^4$	8.377	8.4181	−0.5	93.630 k	93.105 k	+0.56	4.95 h	10.49 s
85	At	[Xe] $4f^{14}\,5d^{10}\,6s^2\,6p^5$	9.617	9.3175	+3.2	96.277 k	95.730 k	+0.57	5.08 h	12.55 s
86	Rn	[Xe] $4f^{14}\,5d^{10}\,6s^2\,6p^6$	10.841	10.7485	+0.9	98.974 k	98.404 k	+0.58	5.22 h	14.67 s
87	Fr	[Rn] $7s^1$	4.305	4.0727	+5.7	101.730 k	101.137 k	+0.59	5.36 h	3.23 s
88	Ra	[Rn] $7s^2$	5.496	5.2784	+4.1	104.541 k	103.922 k	+0.60	5.50 h	4.59 s
89	Ac	[Rn] $6d^1\,7s^2$	5.307	5.3802	−1.4	107.408 k	106.755 k	+0.61	5.65 h	8.20 s
90	Th	[Rn] $6d^2\,7s^2$	6.539	6.3067	+3.7	110.330 k	109.651 k	+0.62	5.80 h	6.53 s
91	Pa	[Rn] $5f^2\,6d^1\,7s^2$	5.741	5.8900†	−2.5	113.302 k	112.601 k	+0.62	5.95 h	9.67 s
92	U	[Rn] $5f^3\,6d^1\,7s^2$	5.820	6.1940	−6.0	116.335 k	115.606 k	+0.63	6.10 h	10.10 s
93	Np	[Rn] $5f^4\,6d^1\,7s^2$	5.856	6.2656	−6.5	119.432 k	–	–	6.26 h	10.43 s
94	Pu	[Rn] $5f^6\,7s^2$	6.108	6.0258	+1.4	122.582 k	–	–	6.42 h	5.84 s
95	Am	[Rn] $5f^7\,7s^2$	6.154	5.9738	+3.0	125.805 k	–	–	6.58 h	5.93 s
96	Cm	[Rn] $5f^7\,6d^1\,7s^2$	5.806	5.9922	−3.1	129.095 k	–	–	6.74 h	11.00 s
97	Bk	[Rn] $5f^9\,7s^2$	6.454	6.1978	+4.1	132.445 k	–	–	6.91 h	6.21 s
98	Cf	[Rn] $5f^{10}\,7s^2$	6.573	6.2819	+4.6	135.866 k	–	–	7.08 h	6.32 s
99	Es	[Rn] $5f^{11}\,7s^2$	6.683	6.3684	+4.9	139.360 k	–	–	7.26 h	6.43 s
100	Fm	[Rn] $5f^{12}\,7s^2$	6.788	6.5000†	+4.4	142.923 k	–	–	7.44 h	6.54 s
101	Md	[Rn] $5f^{13}\,7s^2$	6.890	6.5800†	+4.7	146.564 k	–	–	7.62 h	6.65 s
102	No	[Rn] $5f^{14}\,7s^2$	6.991	6.6262	+5.5	150.282 k	–	–	7.80 h	6.76 s
103	Lr	[Rn] $5f^{14}\,7s^2\,7p^1$	4.844	4.9600	−2.3	154.082 k	–	–	7.99 h	4.78 s
104	Rf	[Rn] $5f^{14}\,6d^2\,7s^2$ *	6.089	6.0200†	+1.2	157.965 k	–	–	8.18 h	12.72 s
105	Db	[Rn] $5f^{14}\,6d^3\,7s^2$ *	7.300	6.8000†	+7.4	161.934 k	–	–	8.38 h	15.74 s
106	Sg	[Rn] $5f^{14}\,6d^4\,7s^2$ *	8.459	7.8000†	+8.4	165.991 k	–	–	8.58 h	18.66 s

Z	El.	configuration	IE calc	IE ref	err %	K calc	K meas	err %	1s clock	valence
107	Bh	[Rn] $5f^{14}\,6d^{5}\,7s^{2}$ *	9.476	7.7000†	+23.1	170.138 k	–	–	8.78 h	12.45 s
108	Hs	[Rn] $5f^{14}\,6d^{6}\,7s^{2}$ *	8.508	7.6000†	+11.9	174.381 k	–	–	8.99 h	24.23 s
109	Mt	[Rn] $5f^{14}\,6d^{7}\,7s^{2}$ *	9.797	8.2914†	+18.2	178.705 k	–	–	9.20 h	26.99 s
110	Ds	[Rn] $5f^{14}\,6d^{8}\,7s^{2}$ *	11.029	9.9497†	+10.8	183.137 k	–	–	9.42 h	29.82 s
111	Rg	[Rn] $5f^{14}\,6d^{9}\,7s^{2}$ *	12.225	10.5716†	+15.6	187.674 k	–	–	9.64 h	32.70 s
112	Cn	[Rn] $5f^{14}\,6d^{10}\,7s^{2}$ *	13.396	11.9707†	+11.9	192.313 k	–	–	9.87 h	35.62 s
113	Nh	[Rn] $5f^{14}\,6d^{10}\,7s^{2}\,7p^{1}$ *	7.603	7.3058†	+4.1	197.070 k	–	–	10.10 h	11.76 s
114	Fl	[Rn] $5f^{14}\,6d^{10}\,7s^{2}\,7p^{2}$ *	8.895	8.6251†	+3.1	201.936 k	–	–	10.33 h	14.07 s
115	Mc	[Rn] $5f^{14}\,6d^{10}\,7s^{2}\,7p^{3}$ *	5.573	5.5791†	−0.1	206.927 k	–	–	10.57 h	7.27 s
116	Lv	[Rn] $5f^{14}\,6d^{10}\,7s^{2}\,7p^{4}$ *	6.831	6.8808†	−0.7	212.036 k	–	–	10.81 h	9.26 s
117	Ts	[Rn] $5f^{14}\,6d^{10}\,7s^{2}\,7p^{5}$ *	8.001	7.6996†	+3.9	217.278 k	–	–	11.06 h	11.26 s
118	Og	[Rn] $5f^{14}\,6d^{10}\,7s^{2}\,7p^{6}$ *	9.127	8.9133†	+2.4	222.656 k	–	–	11.32 h	13.30 s

F Alpha emitters

Gamow half-lives use $r_0 = 1.2$ fm and no fitted constant. Measured values marked † are alpha partial half-lives.

Erratum (*). The second edition listed U-238 with $Q = 4.271$ MeV. AME2020 atomic masses give $Q = 4.2699$ MeV, about 1 keV lower. With the corrected value the Gamow half-life is 4.69×10^9 yr, not 4.58×10^9 yr, and the log ratio is +0.02, not +0.01 (worked step by step in Section 3.5). Recomputing all 35 rows from AME2020 reproduces every other row's Q, β, η, $2G$ and log ratio to within one unit in the last printed digit, and every other Gamow half-life to within 1.5% (under 0.01 decades). The summary statistics move slightly: the RMS error at $r_0 = 1.2$ fm is 0.553 decades (printed 0.55; the second edition printed 0.56), and at $r_0 = 1.15$ fm it is 0.457 (printed 0.46; previously 0.45).

nuclide	Q (MeV)	β	η	$2G$	Gamow $t_{1/2}$	measured	$\log_{10}$ ratio
Te-106	4.290	0.0489	14.92	39.5	97.7 µs	78 µs	+0.10
Nd-144	1.901	0.0324	26.13	101.6	4.74×10^{15} yr	2.29×10^{15} yr	+0.32
Sm-146	2.529	0.0374	23.44	83.8	7.83×10^{7} yr	9.2×10^{7} yr	−0.07
Sm-148	1.987	0.0331	26.45	102.3	9.65×10^{15} yr	6.3×10^{15} yr	+0.19
Gd-148	3.271	0.0425	21.30	69.5	43.6 yr	86.9 yr	−0.30
Dy-154	2.945	0.0403	23.19	79.7	1.29×10^{6} yr	1.4×10^{6} yr	−0.03
Pt-190	3.269	0.0423	26.21	90.5	6.32×10^{10} yr	4.83×10^{11} yr	−0.88
Po-210	5.408	0.0544	22.00	61.2	3.4 d	138 d	−1.61
Po-212	8.954	0.0700	17.10	32.3	0.0629 µs	0.294 µs	−0.67
Po-214	7.834	0.0654	18.29	39.0	55.4 µs	163 µs	−0.47
Po-216	6.906	0.0614	19.48	45.9	56.8 ms	144 ms	−0.40
Po-218	6.115	0.0578	20.70	53.0	77.6 s	3.1 min	−0.38
Rn-220	6.405	0.0592	20.72	52.2	33.6 s	55.6 s	−0.22
Rn-222	5.590	0.0553	22.18	60.8	2.33 d	3.82 d	−0.21
Ra-224	5.789	0.0562	22.32	60.7	2.02 d	3.63 d	−0.25
Ra-226	4.871	0.0516	24.33	72.8	1.05×10^{3} yr	1.6×10^{3} yr	−0.18
Th-228	5.520	0.0549	23.39	66.1	1.27 yr	1.91 yr	−0.18
Th-230	4.770	0.0510	25.16	76.8	5.78×10^{4} yr	7.54×10^{4} yr	−0.12
Th-232	4.082	0.0472	27.21	89.1	1.47×10^{10} yr	1.4×10^{10} yr	+0.02
U-234	4.857	0.0515	25.51	77.8	1.62×10^{5} yr	2.46×10^{5} yr	−0.18
U-236	4.573	0.0500	26.29	82.5	1.79×10^{7} yr	2.34×10^{7} yr	−0.12
U-238*	4.270	0.0483	27.21	88.0	4.69×10^{9} yr	4.46×10^{9} yr	+0.02
Pu-238	5.593	0.0552	24.30	69.5	36.3 yr	87.7 yr	−0.38
Pu-240	5.256	0.0536	25.07	74.0	3.59×10^{3} yr	6.56×10^{3} yr	−0.26
Pu-242	4.984	0.0521	25.75	78.0	2.03×10^{5} yr	3.75×10^{5} yr	−0.27
Pu-244	4.666	0.0504	26.61	83.2	3.74×10^{7} yr	8.14×10^{7} yr†	−0.34
Cm-244	5.902	0.0567	24.18	67.6	5.54 yr	18.1 yr	−0.51
Cm-248	5.162	0.0531	25.86	77.6	1.26×10^{5} yr	3.8×10^{5} yr†	−0.48

nuclide	Q (MeV)	β	η	$2G$	Gamow $t_{1/2}$	measured	$\log_{10}$ ratio
Cf-252	6.217	0.0582	24.07	65.7	307 d	2.73 yr†	−0.51
Fm-254	7.307	0.0631	22.66	56.5	38.1 min	3.24 h	−0.71
No-254	8.226	0.0670	21.79	50.6	5.89 s	56.9 s†	−0.99
Hs-270	9.070	0.0703	22.01	48.8	1.04 s	7.6 s	−0.86
Fl-288	10.076	0.0740	22.08	46.3	75.8 ms	653 ms	−0.94
Lv-292	10.791	0.0766	21.71	43.3	3.81 ms	16 ms	−0.62
Og-294	11.867	0.0803	21.07	38.9	42.6 μs	580 μs	−1.13

G The radioactive edge: longest-lived isotopes

NUBASE2020 values as quoted on the isotope pages. Bracketed values are alternatives printed there.

Z	isotope	half-life	mode
43	Tc-97	$4.21(16)\times10^{6}$ y	EC
61	Pm-145	17.7(4) y	EC
83	Bi-209	$2.01(8)\times10^{19}$ y	α
84	Po-209	124(3) y	α
85	At-210	8.1(4) h	EC/β^{+}
86	Rn-222	3.8215(2) d	α
87	Fr-223	22.00(7) min	β^{-}
88	Ra-226	1600(7) y	α
89	Ac-227	21.772(3) y	β^{-}
90	Th-232	$1.40(1)\times10^{10}$ y	α
91	Pa-231	$3.265(20)\times10^{4}$ y	α
92	U-238	$4.463(3)\times10^{9}$ y	α
93	Np-237	$2.144(7)\times10^{6}$ y	α
94	Pu-244	$8.13(3)\times10^{7}$ y	α
95	Am-243	7350(9) y	α
96	Cm-247	$1.56(5)\times10^{7}$ y	α
97	Bk-247	$1.38(25)\times10^{3}$ y	α
98	Cf-251	898(44) y	α
99	Es-252	471.7(19) d	α
100	Fm-257	100.5(2) d	α
101	Md-258	51.59(29) d	α
102	No-259	58(5) min	α
103	Lr-266	11 (+21/−5) h [22(14) h]	SF
104	Rf-267	48 (+23/−12) min	SF
105	Db-268	16 (+6/−4) h	α/SF
106	Sg-269	13 (+6/−4) min	α
107	Bh-270	2.4 (+4.4/−0.9) min	α
108	Hs-271	46 (+56/−16) s	α
109	Mt-278	4.5 (+3.5/−1.3) s [6(3) s]	α
110	Ds-281	14(3) s	SF
111	Rg-282	100 (+70/−30) s	α
112	Cn-285	28 (+9/−6) s [30(8) s]	α
113	Nh-286	9.5 (+6.3/−2.7) s [12(5) s]	α
114	Fl-289	1.9 (+0.7/−0.4) s [2.1(6) s]	α
115	Mc-290	650 (+490/−200) ms	α
116	Lv-293	57 (+43/−17) ms [70(30) ms]	α
117	Ts-294	51 (+38/−16) ms [70(30) ms]	α
118	Og-294	0.58 (+0.44/−0.18) ms [0.7(3) ms]	α

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